



CREDENTIAL
FOR SOURCE APPROVAL

(Manufacturer of Elastomeric Bearing)

Address : A-1010, Kanakia Wall Street
Andheri - Kurla Road, Chakala
Andheri (E), Mumbai - 400 093



00 91 8976924271 / 8975767254



**E – Mail – polymerproduct@rediffmail.com
polymerproducts09@gmail.com**



Website – www.dynamicprestress.org

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INTRODUCTION

INTRODUCTION

Dynamic Group is a 45 year old well established company in India providing specialise services for pre stressing work of Nuclear containment structure, Bridges, Fly overs, Silos, Metro etc.

Dynamic Group has excellent experience in design, manufacturing, supply and installation of Elastomeric & POT-PTFE bridge bearings.

We have been supplying bearings for Manufacturing, Testing for several projects of Indian Railways, National Highways, Public Works Departments, Metros, Irrigation Projects etc.

The company is providing the bearings as per Indian & international standards to the contractors.

During last decade, the company has registered phenomenal growth in terms of turnover, profitability, production capacity, equipment and client base etc.

NEOPRENE BEARING

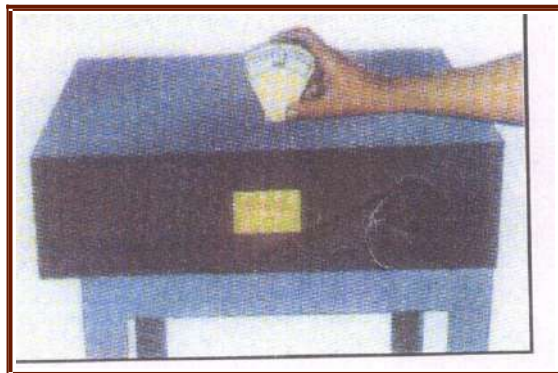
Neoprene is a vital part of any bridge structure or super structure like Aquaduct, Stadium Tunnels, Atomic Reactors, Thermal power stations etc. The primary function of bridge bearings is the accommodation of expansion and contraction due to change in temperature and humidity and to lesser friction cause by traffic loads upon superstructure of the bridge.

Dynamic Neoprene Bearings is manufactured at the state of art manufacturing unit and has In house testing facility and required load that is needed to take care out stringent quality control for satisfaction of performance to the users which guarantee the quality.

We are also been pre qualified for supply of Neoprene bearing by MORTH, RDSO & MSRDC

Advantage of Neoprene Bearings:

- Strong
- Hard: still flexible
- Weatherproof
- Economical
- Durable
- Maintenance Free





ISO & CE CERTIFICATE



CERTIFICATE

**Management system as per
EN ISO 9001:2015**

In accordance with TÜV AUSTRIA procedures, it is hereby certified that

POLYMER PRODUCTS (Plant IV)

**H - 32, MIDC, Satpur, Nashik - 422007,
Maharashtra, India**

applies a management system in line with the above standard for the following scope

**Manufacturing of Elastomeric Bearings, Neoprene Pad with PTFE Sheet,
Neoprene Sheet/ Pad & Seismic Pads**

Certificate Registration No. 20100183004047 / 04

Valid until: 2027-07-19

Certification Body
at TÜV AUSTRIA GMBH

Vienna, 2024-09-02

This certification was conducted in accordance with TÜV AUSTRIA auditing and certification procedures and is subject to regular surveillance audits. The certificate is valid in conjunction with the main certificate.

TÜV AUSTRIA GMBH Deutschstraße 10 A-1230 Wien www.tuv.at



CE CERTIFICATE

Certificate of Compliance



We Hereby Declare That the Technical File of Product Complied With the Requirement of Machinery Directive 2006/42/EC

DYNAMIC PRESTRESS (I) PVT LTD.

Address: F-126, MIDC, SATPUR, NASHIK, MAHARASHTRA (INDIA) PIN 422 007

Product: "MANUFACTURING OF PRESTRESSING ANCHORAGES, STRAND COUPLER, SWAGED ANCHORAGES, REBAR COUPLER, HDPE DUCT, PP DUCT, PRESTRESSING EQUIPMENT, GROUTING EQUIPMENT, POT - PTFE BEARING, SPHERICAL BEARING, ELASTOMERIC BEARING, WELL SINKING JACKS, BAR JACKS, LIFTING & SHIFTING JACKS, VOID FORMER, GROUT CAPS, BARREL & WEDGES."

Certification body has performed an audit of the above product quality system covering the design, manufacture and final inspection of certified product. The quality system has been assessed, approved and is subject to Continues surveillance according to 2006/42/EC Machinery Directive.

This certificate issued under the following conditions:

- 1- It applies only to the quality system maintained in the manufacture of above revered models and it does Not substitute, the design or type examination procedures, if requested,
- 2- The certificate remains valid until the Manufacturing conditions or the quality system are changed,
- 3- The certificate validity is conditioned by positive results of surveillance audits,
- 4- After fulfilling relevant EU legislation, the manufacturer shall affix to each device, of the revered models The CE mark as shown above can be used under. The responsibility of the manufacturer after completion of an EC
- 5- Declaration of conformity and compliance with all relevant EC Directives. The statement is based on A single evaluation of one sample of above mentioned Products. Not imply an assessment of the whole Production.

Date of Certification: 16/01/2026

2nd Surveillance Due: 15/01/2028

1st Surveillance Due: 15/01/2027

Certificate Expiry: 15/01/2029

(The company, maintaining its system to the required standard)

Certificate Number: CE-26250

Validity of this certificate can be verified at www.ukiac.org.uk



Validity of This Certificate is Successful. Annual Surveillance Audit to Be Done.
This Certificate is the Property of UKIAC. Shall Be Returned Immediately Upon Request, MS.CERT is accredited by
UKIAC (www.ukiac.org.uk), 29 Constance street London England E16
E-mail: info@ukiac.org.uk

RDSO
APPROVAL



भारत सरकार - रेल मंत्रालय
अनुसंधान अधिकार्य और मानक संगठन
लुक्नो - 226 011
EPBX (0522) 2451200
Fax (0522) 2458500

Government of India-Ministry of Railways
Research Designs & Standards Organisation
Lucknow - 226 011
DID (0522) 2450116
DID (0522) 2460310



Date: 06.07.2024

No: CBS/PBEJ/Reg/Polymer

M/s. Polymer Products

Mumbai

Email: polymerproduct@rediffmail.com

Office Address	Work address
72, Mahavir Centre, Sector-17, Vashi, Navi Mumbai-400705	H-32, MIDC, Satpur, Nasik-422007

Sub: **Quality Audit for manufacture and supply of Elastomeric Bearing (Item ID 3100417001)**

1. Quality Audit of M/s. Polymer Products, Nasik conducted successfully on **26.06.2024** at their work address **H-32, MIDC, Satpur, Nasik-422007** for manufacture and supply of following item-

SN	Item	Specification
(i)	Manufacture and supply of Elastomeric Bearing (Item ID 3100417001)	As mentioned in STR No. BS-S-7.5.3.1-10 (Version 1.1)

2. Next Quality Audit: to be completed before five years i.e. **25.06.2029**

Digitally Signed by Subhash
Chandra

Date: 08-07-2024 09:32:23

Reason: Approved

(Subhash Chandra)

Director/ B&S-III

For Director General (B&S)

Copy to : All concerned as per Mailing list enclosed:

MAILING LIST

I. Chief Bridge Engineer

1. Central Railway, Mumbai CST - 400 001
2. Eastern Railway, Fairlie Place, Kolkata-700 001
3. Northern Railway, Baroda House, New Delhi-110 001.
4. N.E.Railway, Gorakhpur-273 012
5. Southern Railway, Park Town, Chennai-600 003
6. S.C.Railway, Rail Nilayam, Secunderabad-500 371
7. Western Railway, Churchgate, Mumbai-400 020.
8. South Eastern Railway, Garden Reach, Kolkata-700 043
9. N.F.Railway, Maligaon, Guwahati – 781 011
10. East Central Railway, Hajipur – 844 101
11. East Coast Railway, Bhubaneshwar – 364 001
12. North Central Railway, HQ office, Subedarganj, Block-Bhagirathi, Prayagraj – 211 015
13. North Western Railway, Jaipur – 755 001
14. South East Central Railway, Bilaspur-495 004
15. South Western Railway, Hubli – 580 023
16. West Central Railway, Jabalpur – 482 001

II.The Chief Administrative Officer (Const.)

1. Central Railway, New Administrative Office Building, 6th Floor, D.N. Road, Mumbai CST-400001
2. CAO(I), Eastern Railway, 4th Floor, New Koilaghat Building, Strand Road, Kolkata- 700001
3. CAO(II), Eastern Railway, 4th Floor, New Koilaghat Building, Strand Road, Kolkata- 700001
4. Northern Railway, Kashmere Gate, Delhi – 110 001
5. USBRL Project, Opp. R.R.B., Jammu Tawi - 180013
6. North Eastern Railway, Gorakhpur – 273012.
7. Southern Railway, Egmore, Chennai – 600 003
8. West Central Railway, Jabalpur - 482001
9. South Western Railway, 18, Millar Road, Bangalore – 560 046
10. South Central Railway, Rail Nilayam, Secunderabad – 500371
11. South Eastern Railway, Garden Reach, Kolkata – 700043
12. East Coast Railway, Chandrashekharpur– 751023, Bhubaneshwar
13. Western Railway, Churchgate-400020
14. CAO(North) , East Central Railway, Mahendrugat, Patna- 80004
15. CAO(South) , East Central Railway, Mahendrugat, Patna - 800004
16. North Western Railway, Near Railway Station, Jaipur-302001
17. South East Central Railway, Bilaspur - 495004 (Chhattisgarh)
18. North Central Railway, HQ office, Subedarganj, Block-Bhagirathi, Prayagraj – 211 015
19. CAO(I), N.F. Railway, Maligaon Guwahati – 781001
20. CAO(II), N.F. Railway, Maligaon, Guwahati – 781001

III.Chairman-cum-Managing Directors

1. MD, RITES Ltd., RITES BHAWAN. No. 1, Sector-29, Gurgaon-122001 (Haryana)
2. MD, IRCON International Ltd., C-4, District Centre, Saket, New Delhi-110017
3. CMD/ Konkan Railway Corporation Ltd., BelapurBhawan, Plot No. 6, Sector 11, CBD, Belapur, New Mumbai-400614
4. MD, RVNL, August Plot No. 25, 1st Floor, August KrantiBhawan, BhikajiCama Palace, R.K. Puram, New Delhi- 110066
5. MD, Dedicated Freight Corridor Corporation of India Ltd., Sth Floor, PragatiMaidan, Metro Station Building Complex, New Delhi-110001



सत्यमेव जयते

No: CBS/PBEJ/Reg/ PP

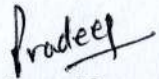
Date: 06.04.2023

M/s. Polymer Products
72-Mahavir Centre, Sector-17
Vashi Navi Mumbai,
Maharashtra - 400705, India

Sub: Compliance of revised STR of Elastomeric bearing.

Ref: Your letter No. PP/RDSO/2022-23/399 Dated 26.05.2022

1. Vide letter under Ref., firm had sent the compliance of revised STR and it was found in order during physical verification by RDSO official. Hence with approval of competent authority, firm is hereby continued in RDSO approved list of Elastomeric bearing.
2. The other conditions given in registration certificate issued vide RDSO letter No. CBS/PBEJ/Reg/ PP dated 29.12.2019 have no change.


(Pradeep Kumar)
Director/B&S/II

DMRC

APPROVED MANUFACTURES/SUPPLIERS For Civil Works

March 2023

All materials and products shall conform to the Outline Construction Specification (OCS) of DMRC, BIS codes and other relevant codes etc. and shall be of make as approved by DMRC.

The list of approved makes for products and materials is given below. No Further approval is required to be taken for usage of these makes. Further, the request of equivalent manufactures shall only be considered as an exception wherein it is certified by all the approved manufacturer's/suppliers that they are not able to provide the required quantity of products/materials.

S. No.	Details of Materials/ Products	Manufacturer's/Supplier's Name
1.	Cement	Ultratech, Orient Cement, JK Cement ,Birla Corporation, Wonder Cement, ,DALMIA Bharat, ACC, Ambuja
2.	Reinforcement Bars	Prequalified Manufacturers SAIL, RINL, TISCO, IISCO, Jindal and manufacturers as per RDSO's latest approved list
3.	Epoxy	FOSROC, Asian paints,SUNANDA, SIKA, FAIRMATE, KunalConchem,Hindcon Chemicals, STP Limited
4(a).	Expansion Joints for Viaduct	Prequalified Manufacturers as per RDSO's latest approved list
4(b).	Expansion Joints for buildings	VR Engineers ,Maurer Sanfield, DECG International
5.	Admixtures	Durabuild, TP Buildtech, Hindcon Chemicals, Vista Chemtech, Buildtech, Rheoplast, Asian Lab, CICO Technologies,Yahska Polymers, FOSROC, Asian Paints, CAC, Ado Additives, Prism Johnson ,STP Limited ,SIKA, Normet India , KunalConchem, Polygon chemicals, FairmateIndia,TREMCO CPG, Pinnacle, Shalimar Paints/Hella Infra, SUNANDA
6(a)	Low Strain Pile Integrity Testing, UPV Test, Rebound Hammer Test	CEG Test House, AADCO Testing, CENGRS, DVG Lab, ADS Labtech (please note that NABL accreditation is mandatory)



6(b).	Cross Hole Testing of Piles	CEG Test House, AADCO Testing, CENGRS, ADS Labtech (please note that NABL accreditation is mandatory)
7.	Anchor Fastener	TRUTEK, Pioneer nuts and bolts, FOSROC, HILTI, Styrene Packings(Please note that ETA Certification is mandatory for using/supplying fasteners for load bearing structural members)
8.	Structural Steel	KL Steels and Sons, TATA, SAIL, ESSAR, Maharashtra Pipes, Jindal Steel & Power Ltd., Steel Works & Power Engineers, SKS Ispat & Power, Shamli Steel, Topworth, Goodluck India, Rimjhim, APL Apollo tubes
9.	Pre- stressing Strand(LRPC)	TATA SSL Ltd., USHA Martin, DP Wires, Miki Steel, Kataria Group
10(a)	Pot/PTFE Bearings, Elastomeric	Deevin Systems, VSIL, Maurer Sanfield, Dynamic Prestress, Kanta Rubber(RDSO approval is mandatory)
10(b)	Elastomeric Bearings	Deevin Systems, VSIL, Maurer Sanfield, Kanta Rubber, Polymer Products, DECG International(RDSO approval is mandatory)
10(c)	Spherical Bearings	Maurer Sanfield, MAGEBA
11.	Horizontal Tie Bars/ Shear Bars	DEXTRA
12.	HDPE Sheathing	Dynamic Prestress, JK Prestressing, FOSROC
13.	Formwork Release Agent	Eastern Petroleum, Yahska polymers, KunalConchem, ADO Additives, STP Limited, FAIRMATE
14.	Prestressing System	Dynamic Prestress, TENSA INDIA, JK Prestressing, Usha Martin
15.	Reinforcement Couplers (cold forged parallel threads type only)	TATA Steel, Maurer Sanfield, DEXTRA India, LEVIAT India, REBAR Couplers India (P) Ltd.
16.	Hollow Sections, Pipes	Surya Roshni, Dadu Pipes, APL Apollo Tubes
17.	Drainage Pipes	TEXMO Pipes
18.	Non Shrink Grout	DURABUILD, Yahska polymers, FOSROC, ASIAN Paints, SIKA India, KunalConchem, FAIRMATE, STP Limited
19.	Bonding Coat	Asian paints, STP Limited, FAIRMATE
20.	Polysulphide Sealant	Asian paints, SIKA, KunalConchem, IWL India, HindconChemicals, STP Limited, FAIRMATE
21.	Steel Structural Fasteners	PANCHSHEEL Fasteners, PIONEER nuts and bolts, KIRTI Fasteners, HARYANA fasteners, IMPERIAL bolts and Fasteners, KARAMTARA (Please note that ETA Certification is mandatory for using/supplying fasteners for load bearing structural members)

Mahesh Sreen

**QUALITY
ASSURANCE
PLAN**

QUALITY ASSURANCE PLAN

Sr. No	Component to be tested	Test to be conducted	No. of samples	Reference Documents	Acceptance Criteria	Testing Agency
1	Rubber Component	1) Tensile strength 2) % Elongation 3) Hardness 4) Adhesion strength	01 No. per lot	IRC : 83 Part (II) – 1987 specification	IRC : 83 Part (II) – 1987 Specification	Manufacturer's Laboratory
2	M.S. Plate	1) Tensile strength 2) % Elongation	One piece per lot of procurement	IRC : 83 Part (II) – 1987 specification	IRC : 83 Part (II) – 1987 specification	Independent Laboratory
3	Testing on finished Bearing	1) Visual inspection 2) Dimension check 3) Hardness 4) Elastic module check 5) Shear modulus test 6) Stripping strength test 7) Ultimate compressive test	All bearings One piece per order (Above 24 Nos.)	IRC : 83 Part (II) – 1987 specification	IRC : 83 Part (II) – 1987 specification	Manufacturer's Laboratory
1	Rubber Component	1) Compression Test 2) Ageing Test	One No. per Lot	IRC : 83 Part (II) – 1987 specification	IRC : 83 Part (II) – 1987 specification	Manufacturer's Laboratory
2	Testing on finished Bearing	Axial load 5 Mpa and 15 Mpa	All Bearings	IRC : 83 Part (II) – 1987 specification	IRC : 83 Part (II) – 1987 specification	Manufacturer's Laboratory

IN HOUSE

&

THIRD PARTY

TEST REPORTS

Corporate Office :
72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.
Tel. : +91 22 2789 2628/1310
Website : www.dynamicprestress.com

Works :
H - 32, M.I.D.C., Satpur, Nashik-422 007.
Tele : (0253) 2350935, 6604461
e-mail : polymerproduct@rediffmail.com

POLYMER



PRODUCTS



TEST REPORT ON ELASTOMERIC BEARINGS

TESTED AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

Report No. : 0307/2223 Date : 10-02-2023
Name of the party : M/s. STRUCT REHAB
Name of the Project : Lifting & Pushing of Span at Km. 936+910 LHS Minor Bridge in our project section Diwancheruvu - Siddhantham Section (from Km.901+500 to Km 950+542) of NH-16 in the State of Andhra Pradesh -- Requirement for Elastomeric Bearings -- Change in Size and Quantity
Bearing Size (mm) : 400 X 200 X 40 mm Quantity : 20 Nos.
Side cover (mm) : 6 mm
Effective Plan Bearing Size (mm) : 388 X 188 mm (Bearing Size - Side cover)
Effective Plan Area of the Bearing (A_1) (cm^2) : 729.4 cm^2 (388 X 188 / 100) (Effective length X width)
Plate Size (mm) : 388 X 188 X 3 mm No. of Plate: 3 Nos.
Thickness of Internal Layer (t_e) : 10 mm No. of Internal Layer : 2 Nos.
Top & Bottom Layer (mm) : 5.5 mm Layer No.: 2 Nos.
Thickness of Elastomer (mm) (T) : 31 mm (10 x 2) + (5.5 x 2)
Area of Vertical Machine ram (a) : 2428.0 cm^2 ($\pi/4 \times D^2$) D = 55.600 cm
Area of Horizontal Machine ram (b) : 324.0 cm^2 ($\pi/4 \times d^2$) d = 20.311 cm

$$\text{Shape Factor (S)} = \frac{(A_1 / l_p \times t_e)}{\text{PERIMETER OF BEARING X 1 LAYER OF RUBBER THICKNESS}}$$

$$\text{Shape Factor (S)} = \{(388 \times 188) / (2 \times (388+188) \times 10)\} = \boxed{6.33} \quad (\text{C.5.1.3.1})$$

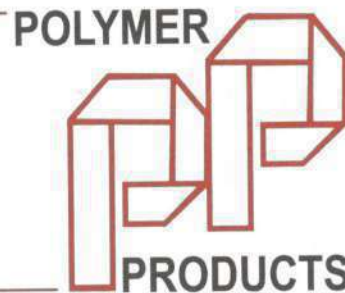
OBSERVATIONS :

TYPE OF TEST	TEST RESULT	PERMISSIBLE RANGE
Hardness (IRHD) (Avg.)	60	60 $\pm$ 5
Compressive Stiffness [Ea] (MPa)	244.36	250 MPa $\pm$ 20 % (I.e. 200 $\leq$ [Ea] $\leq$ 300)
Shear Modulus [G] (MPa)	0.91	0.9 MPa $\pm$ 0.15 (I.e. 0.75 $\leq$ [G] $\leq$ 1.05)
Shear Bond Test (Destructive Test)	Found Ok	No Sign of Bond failure
Destructive Test	Found Ok	M.S. PLATE THICKNESS, INTERNAL LAYER AND TOP AND BOTTOM LAYER AS PER DRAWING AND FOUND OK

Tested by
POLYMER PRODUCTS

TAUSIF KHAN PATHAN
Design Engineer

Corporate Office :
72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.
Tel. : +91 22 2789 2628/1310
Website : www.dynamicprestress.com



Works :
H - 32, M.I.D.C., Satpur, Nashik-422 007.
Tele : (0253) 2350935, 6604461
e-mail : polymerproduct@rediffmail.com

TEST REPORT ON ELASTOMERIC BEARINGS

TESTED AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

Report No.

: 0308/2223

Date : 01-03-2023

Name of the party

: M/s. SHELKE CONSTRUCTION PVT. LTD.

Name of the Project

: REHABILITATION & UP-GRADATION TO FOUR LANING
CONFIGURATION OF THE HIGHWAY STARTING FROM JUNCTION
WITH NH-65 NEAR PATAS, VASUNDE PHATA, UNDEWADI KADE
PATHAR NH-965G AND PARTLY BARAMATI BYPASS IN THE STATE
OF MAHARASHTRA ON EPC MODE. (FROM KM 0/000 TO KM 41/369,
TOTAL LENGTH = 41.369 KM)

Bearing Size (mm)

: **400 X 400 X 76 mm** Quantity : 20 Nos.

Side cover (mm)

: 5 mm

Effective Plan Bearing Size (mm)

: 390 X 390 mm (Bearing Size - Side cover)

Effective Plan Area of the Bearing (A_1) (cm^2)

: 1521.0 cm^2 (390 X 390 / 100) (Effective length X width)

Plate Size (mm)

: 390 X 390 X 4 mm No. of Plate: 5 Nos.

Thickness of Internal Layer (t_e)

: 10 mm No. of Internal Layer : 4 Nos.

Top & Bottom Layer (mm)

: 8 mm Layer No.: 2 Nos.

Thickness of Elastomer (mm) (T)

: 56 mm (10 x 4) + (8 x 2)

Area of Vertical Machine ram (a)

: 2428.0 cm^2 ($\pi/4 \times D^2$) D = 55.600 cm

Area of Horizontal Machine ram (b)

: 324.0 cm^2 ($\pi/4 \times d^2$) d = 20.311 cm

Effective Plan Area of the Bearing (A_1)

Shape Factor (S) = ($A_1 / l_p \times t_e$) = PERIMETER OF BEARING X 1 LAYER OF RUBBER THICKNESS

Shape Factor (S) = $\{(390 \times 390) / (2 \times (390+390) \times 10)\} = \boxed{9.75}$
(C.5.1.3.1)

OBSERVATIONS :

TYPE OF TEST	TEST RESULT	PERMISSIBLE RANGE
Hardness (IRHD) (Avg.)	60	60 $\pm$ 5
Compressive Stiffness [Ea] (MPa)	334.56	350 MPa $\pm$ 20 % (I.e. 280 $\leq$ [Ea] $\leq$ 420)
Shear Modulus [G] (MPa)	0.93	0.9 MPa $\pm$ 0.15 (I.e. 0.75 $\leq$ [G] $\leq$ 1.05)
Shear Bond Test (Destructive Test)	Found Ok	No Sign of Bond failure
Destructive Test	Found Ok	M.S. PLATE THICKNESS, INTERNAL LAYER AND TOP AND BOTTOM LAYER AS PER DRAWING AND FOUND OK

Tested by :

POLYMER PRODUCTS

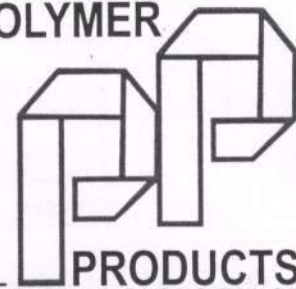
TAUSIF KHAN PATHAN
Design Engineer

Corporate Office :
72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.
Tel. : +91 22 2789 2628/1310
Website : www.dynamicprestress.com

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Tele : (0253) 2350935, 6604461
e-mail : polymerproduct@rediffmail.com

POLYMER



PRODUCTS



TEST REPORT ON ELASTOMERIC BEARINGS

TESTED AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

Report No.

: 0255/2223

Date : 11-01-2023 TO 12-01-2023

Name of the party

: M/s. DMIA-GHV (INDIA) PRIVATE LIMITED.

Name of the Project

: CONSTRUCTION OF EIGHT LANE CARRIAGWAY STARTING NEAR JUNCTION WITH NH-76 TO JUNCTION WITH NH-12 NEAR UMMEDPURA (KM 388.420 TO 419.000) SECTION OF DELHI - VADODARA ACCESS CONTROLLED GREEN FIELD ALIGNMENT (NH-148N) ON EPC MODE UNDER BHARATMALA PARIYOJANA IN THE STATE OF RAJASTHAN

Bearing Size (mm)

: **550 DIA X 156 mm** Quantity : 24+2 Nos

Side cover (mm)

: 8 mm

Effective Plan Bearing Size (mm)

: 534 DIA mm (Bearing Size - Side cover)

Effective Plan Area of the Bearing (A_1) (cm^2)

: 2240.0 cm^2 (534 / 100) (Effective Dia)

Plate Size (mm)

: 534 DIA X 4 mm No. of Plate: 7 Nos

Thickness of Internal Layer (t_e)

: 16 mm No. of Internal Layer : 6 Nos

Top & Bottom Layer (mm)

: 16 mm Layer No.: 2 Nos

Thickness of Elastomer (mm) (T)

: 128 mm (16 x 6) + (16 x 2)

Area of Vertical Machine ram (a)

: 2428.0 cm^2 ($\pi/4 \times D^2$) D = 55.600 cm

Area of Horizontal Machine ram (b)

: 324.0 cm^2 ($\pi/4 \times d^2$) d = 20.311 cm

Shape Factor (S) = ($A_1 / l_p \times t_e$) = $\frac{\text{Effective Plan Area of the Bearing } (A_1)}{\text{PERIMETER OF BEARING X 1 LAYER OF RUBBER THICKNESS}}$

Shape Factor (S) = $\{ (\pi/4) \times (534 \times 534) / [(\pi \times 534) \times 16] \} = \boxed{8.34}$
(C.5.1.3.1)

OBSERVATIONS :

TYPE OF TEST	TEST RESULT	PERMISSIBLE RANGE
Hardness (IRHD) (Avg.)	69	70 ± 5
Compressive Stiffness [Ea] (MPa)	367.94	350 MPa ± 20 % (i.e. 280 ≤ [Ea] ≤ 420)
Shear Modulus [G] (MPa)	1.21	1.15 MPa ± 0.20 (i.e. 0.95 ≤ [G] ≤ 1.35)
Shear Bond Test (Destructive Test)	Found Ok	No Sign of Bond failure
Destructive Test	Found Ok	M.S. PLATE THICKNESS, INTERNAL LAYER AND TOP AND BOTTOM LAYER AS PER DRAWING AND FOUND OK

Tested by :

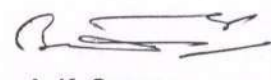
POLYMER PRODUCTS

Witnessed by:


Girish Darandale
General Manager


Sandeep Shah
Structure Engineer
DMIA-GHV
DVE-PKG-14


R. K. Sharma
Deputy General Manager
DMIA - GHV
DVE - PKG-14

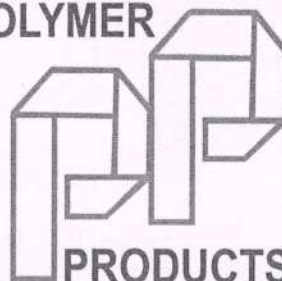

A. K. Saxena
Resident cum Highway Engineer
L N MALVIYA INFRAPROJECTS
DVE PKG-14

Corporate Office :
72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.
Tel. : +91 22 2789 2628/1310
Website : www.dynamicprestress.com

Works :

H - 32, M.I.D.C., Satpur, Nashik-422 007.
Tele : (0253) 2350935, 6604461
e-mail : polymerproduct@rediffmail.com

POLYMER



PRODUCTS



TEST REPORT ON ELASTOMERIC BEARINGS

TESTED AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

Report No.

: 0264/2223

Date : 23-01-2023

Name of the party

: M/s. GANGAMAI INDUSTRIES & CONSTRUCTIONS LIMITED

Name of the Project

: FOUR LANING OF AUSA - CHAKUR OF SECTION OF NH-361 FROM KM.55.835 TO KM. 114.345 (DESIGN LENGTH 58.51) UNDER BHARATMALA PRIYOJNA ON HYBRID ANNUITY MODE IN THE STATE OF MAHARASHTRA.

Bearing Size (mm)

: 450 X 300 X 74 mm

Quantity : 32+2 Nos.

Side cover (mm)

: 6 mm

Effective Plan Bearing Size (mm)

: 438 X 288 mm

(Bearing Size - Side cover)

Effective Plan Area of the Bearing (A_1) (cm^2)

: 1261.4 cm^2

(438 X 288 / 100) (Effective length X width)

Plate Size (mm)

: 438 X 288 X 4 mm

No. of Plate: 5 Nos.

Thickness of Internal Layer (t_e)

: 12 mm

No. of Internal Layer : 4 Nos.

Top & Bottom Layer (mm)

: 3 mm

Layer No.: 2 Nos.

Thickness of Elastomer (mm) (T)

: 54 mm

(12 x 4) + (3 x 2)

Area of Vertical Machine ram (a)

: 2428.0 cm^2

($\pi/4 \times D^2$) D = 55.600 cm

Area of Horizontal Machine ram (b)

: 324.0 cm^2

($\pi/4 \times d^2$) d = 20.311 cm

Shape Factor (S) = ($A_1 / l_p \times t_e$) = Effective Plan Area of the Bearing (A_1)
PERIMETER OF BEARING X 1 LAYER OF RUBBER THICKNESS

Shape Factor (S) = {(438 x 288) / (2 x (438+288) x 12)} = **7.24**
(C.5.1.3.1)

OBSERVATIONS :

TYPE OF TEST	TEST RESULT	PERMISSIBLE RANGE
Hardness (IRHD) (Avg.)	61	60 $\pm$ 5
Compressive Stiffness [Ea] (MPa)	172.19	150 MPa $\pm$ 20 % (I.e. 120 $\leq$ [Ea] $\leq$ 180)
Shear Modulus [G] (MPa)	0.99	0.9 MPa $\pm$ 0.15 (I.e. 0.75 $\leq$ [G] $\leq$ 1.05)
Shear Bond Test (Destructive Test)	Found Ok	No Sign of Bond failure
Destructive Test	Found Ok	M.S. PLATE THICKNESS, INTERNAL LAYER AND TOP AND BOTTOM LAYER AS PER DRAWING AND FOUND OK

Tested by :

POLYMER PRODUCTS

Witnessed by:

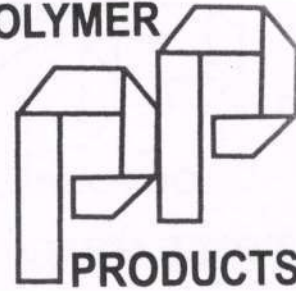
Girish Darandale
General Manager

Suraj Chavan
Deputy Project Manager
Gangamai Industries & constructions Limited

Eshwar Kumar Bhukya
Bridge Engineer
SA INFRA

Corporate Office :

A-1010, Kanakia Wall Street, Andheri-Kurla Road,
Chakala, Andheri East, Mumbai - 400 093
Tel. : 8976924271
Website : www.dynamicprestress.com

POLYMER**Works :**

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TEST REPORT ON ELASTOMERIC BEARINGS

TESTED AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

Report No.

: 0330/2223

Date : 27-03-2023**Name of the party**

: M/s. KALYAN TOLL INFRASTRUCTURE LTD

Name of the Project

: FOUR LANNING OF CHAKUR - LOHA SECTION OF NH-361
FROM KM. 114.600 TO KM. 187.800 (EXISTING CHAINAGE KM.
528.200 ON MSH-3 TO KM.514.885 ON MSH-2 UNDER NH(O) ON
HYBRID ANNUITY MODE IN THE STATE OF MAHARASHTRA.

Bearing Size (mm): **500 X 450 X 104 mm** **Quantity :** 20 Nos.**Side cover (mm)**

: 6 mm

Effective Plan Bearing Size (mm)

: 488 X 438 mm (Bearing Size - Side cover

Effective Plan Area of the Bearing (A_1) (cm^2): 2137.4 cm^2 (488 X 438 / 100) (Effective length X width**Plate Size (mm)**: 488 X 438 X 3 mm **No. of Plate:** 6 Nos.**Thickness of Internal Layer (t_e)**: 16 mm **No. of Internal Layer :** 5 Nos.**Top & Bottom Layer (mm)**: 3 mm **Layer No.:** 2 Nos.**Thickness of Elastomer (mm) (T)**

: 86 mm (16 x 5) + (3 x 2)

Area of Vertical Machine ram (a): 2428.0 cm^2 ($\pi/4 \times D^2$) D = 55.600 cm**Area of Horizontal Machine ram (b)**: 324.0 cm^2 ($\pi/4 \times d^2$) d = 20.311 cmEffective Plan Area of the Bearing (A_1)

$$\text{Shape Factor (S)} = (A_1 / l_p \times t_e) = \frac{\text{Effective Plan Area of the Bearing (A}_1\text{)}}{\text{PERIMETER OF BEARING X (INNER LAYER LAYER + (1.4 x OUTER LAYER))}}$$

$$\text{Shape Factor (S)} = \{(488 \times 438) / (2 \times (488+438) \times 16 + 1.4 \times 3)\} = \boxed{5.71}$$

(C.5.1.3.1)

OBSERVATIONS :

TYPE OF TEST	TEST RESULT	PERMISSIBLE RANGE
Hardness (IRHD) (Avg.)	60	60 ± 5
Compressive Stiffness [Ea] (MPa)	252.97	250 MPa ± 20 % (I.e. 200 ≤ [Ea] ≤ 300)
Shear Modulus [G] (MPa)	0.91	0.9 MPa ± 0.15 (I.e. 0.75 ≤ [G] ≤ 1.05)

Tested by :

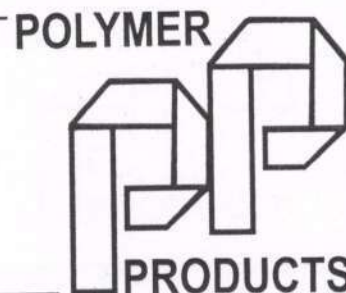
POLYMER PRODUCTS

TAUSIF KHAN PATHAN

Design Engineer

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e-mail : polymerproduct@rediffmail.com



TEST REPORT ON ELASTOMERIC BEARINGS

TESTED AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

Report No.	: 0294/2223	Date : 17-10-2022
Name of the party	: M/s. RACHANA.	
Name of the Project	: CONSTRUCTION Of ROB With Approaches on NH 848-B At Karambeli Railway Crossing L.C No. 77 @ KM 0/0 To 0/600 in the State of Gujarat on EPC Mode Under NH(O)	
Bearing Size (mm)	: 500 X 250 X 65 mm	Quantity : 192+2 Nos.
Side cover (mm)	: 6 mm	
Effective Plan Bearing Size (mm)	: 488 X 238 mm	(Bearing Size - Side cover)
Effective Plan Area of the Bearing (A_1) (cm^2)	: 1161.4 cm^2	(488 X 238 / 100) (Effective length X width)
Plate Size (mm)	: 488 X 238 X 3 mm	No. of Plate: 5 Nos.
Thickness of Internal Layer (t_e)	: 10 mm	No. of Internal Layer : 4 Nos.
Top & Bottom Layer (mm)	: 5 mm	Layer No.: 2 Nos.
Thickness of Elastomer (mm) (T)	: 50 mm	(10 x 4) + (5 x 2)
Area of Vertical Machine ram (a)	: 2428.0 cm^2	($\pi/4 \times D^2$) D = 55.600 cm
Area of Horizontal Machine ram (b)	: 324.0 cm^2	($\pi/4 \times d^2$) d = 20.311 cm

$$\text{Shape Factor (S)} = (A_1 / l_p \times t_e) = \frac{\text{Effective Plan Area of the Bearing (A}_1\text{)}}{\text{PERIMETER OF BEARING X 1 LAYER OF RUBBER THICKNESS}}$$

$$\text{Shape Factor (S)} = \{(488 \times 238) / (2 \times (488+238) \times 10)\} = \boxed{8.00}$$

(C.5.1.3.1)

OBSERVATIONS :

TYPE OF TEST	TEST RESULT	PERMISSIBLE RANGE
Hardness (IRHD) (Avg.)	60	60 $\pm$ 5
Compressive Stiffness [Ea] (MPa)	329.41	350 MPa $\pm$ 20 % (I.e. 280 $\leq$ [Ea] $\leq$ 420)
Shear Modulus [G] (MPa)	0.94	0.9 MPa $\pm$ 0.15 (I.e. 0.75 $\leq$ [G] $\leq$ 1.05)
Shear Bond Test (Destructive Test)	Found Ok	No Sign of Bond failure
Destructive Test	Found Ok	M.S. PLATE THICKNESS, INTERNAL LAYER AND TOP AND BOTTOM LAYER AS PER DRAWING AND FOUND OK

Tested by :

POLYMER PRODUCTS

TAUSIF KHAN PATHAN
Design Engineer

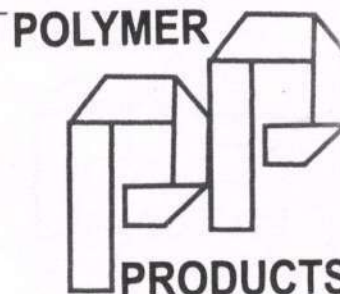
Witnessed by:

SHRAY PATEL
RACHANA

M. A. PATEL
ASSISTANT ENGINEER
NH SUB DIVISION VALSAD

Corporate Office :
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Chakala, Andheri East, Mumbai - 400 093
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TEST REPORT ON ELASTOMERIC BEARINGS

TESTED AS PER IRC:83-2018 (PART-II) SPECIFICATIONS

Report No.
Name of the party
Name of the Project

: 0335/2223 **Date : 28-03-2023**
: M/s. T AND T INFRA LTD.
: IMP. TO NH 361A TO KHUP SARWADI - VISHNUPURI - ASARJAN - NH
361A - BALIRAMPUR - NH 361- TUPPA -
WADGAON - BRAMHANWADA - PATHRAD -
RAILWAYSTATION -DEGAON(BK) - YELEGAON -
AMARABAD - LONI(KH) - RODGI NIMGAON TO NH 361ROAD WITH
KALESHWAR LONG BRIDGE AND LA SH-412 KM 0/0 TO 7/00 AND
14/00 TO 24/00 (ACTUAL WORKING CHAINAGE 5/00 TO 6/080 WITH
ROAD CONNECTING TO PARKING) ON ENGINEERING,
PROCUREMENT & CONSTRUCTION (EPC) MODE

Bearing Size (mm) **: 400 X 400 X 60 mm** **Quantity : 24+2 Nos**
Side cover (mm) **: 4 mm**
Effective Plan Bearing Size (mm) **: 392 X 392 mm** **(Bearing Size - Side cover)**
Effective Plan Area of the Bearing (A₁) (cm²) **: 1536.6 cm²** **(392 X 392 / 100) (Effective length X width)**
Plate Size (mm) **: 392 X 392 X 4 mm** **No. of Plate: 4 Nos**
Thickness of Internal Layer (t_e) **: 12 mm** **No. of Internal Layer : 3 Nos**
Top & Bottom Layer (mm) **: 4 mm** **Layer No.: 2 Nos**
Thickness of Elastomer (mm) (T) **: 44 mm** **(12 x 3) + (4 x 2)**
Area of Vertical Machine ram (a) **: 2428.0 cm²** **(π/4 x D²) D = 55.600 cm**
Area of Horizontal Machine ram (b) **: 324.0 cm²** **(π/4 x d²) d = 20.311 cm**

Shape Factor (S) = (A₁ / l_p x t_e) = **Effective Plan Area of the Bearing (A₁)**
PERIMETER OF BEARING X {INNER LAYER LAYER + (1.4 x OUTER LAYER)}

Shape Factor (S) = {(392 x 392) / (2 x (392+392) x 12 + 1.4 x 4)} = **5.57**
(C.5.1.3.1)

OBSERVATIONS :

TYPE OF TEST	TEST RESULT	PERMISSIBLE RANGE
Hardness (IRHD) (Avg.)	60	60 ± 5
Compressive Stiffness [Ea] (MPa)	249.94	250 MPa ± 20 % (I.e. 200 ≤ [Ea] ≤ 300)
Shear Modulus [G] (MPa)	0.92	0.9 MPa ± 0.15 (I.e. 0.75 ≤ [G] ≤ 1.05)
Shear Bond Test (Destructive Test)	Found Ok	No Sign of Bond failure
Destructive Test	Found Ok	M.S. PLATE THICKNESS, INTERNAL LAYER AND TOP AND BOTTOM LAYER AS PER DRAWING AND FOUND OK

Tested by :

POLYMER PRODUCTS

TAUSIF KHAN PATHAN
Design Engineer



IRMRA



INDIAN RUBBER MANUFACTURERS RESEARCH ASSOCIATION

An Autonomous Institute Under DPIIT, Ministry of Commerce & Industry, Government of India

E-mail : info@irmra.org • Website : www.irmra.org

DEVELOPMENTAL EVALUATION REPORT

ULR No : TC771922000001880	07 Sep 2022	Page 1 of 3
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--- CUSTOMER DETAILS ---

Customer Name & Address :	
M/s.Polymer Products - Nashik H-32 MIDC Satpur Nashik-07 Nashik MH	
Contact Person & Telephone No :	Sunil Kotagi -2350935
Period of Testing:	26 Aug 2022 to 07 Sep 2022
Customer Letter Ref No with Date :	PP/NSK/IRMRA/22-23 Dated 17 Aug 2022
Date of Receipt :	26 Aug 2022
Any Other Information like witness / third party	

--- SAMPLE DETAILS ---

Sample Code / Description	Rubber compound
Product Specification	NIL

--- REPORT DETAILS ---

Description of Report	Page Nos
TC771922000001880 - M & R -	2 of 3
TC771922000001880 - C & I -	3 of 3

Terms and Conditions

1. Sample was not drawn by IRMRA, unless otherwise stated.
2. The result enclosed are only related to the sample submitted to us.
3. The report shall not be reproduced except in full without prior written permission of laboratory.
4. Information contained in this letter / or enclosed reports is/are purely meant for your guidance and should not be used for litigation/evidence/advertisement, without prior permission from laboratory.

Prepared and Checked By



भारतीय रबड़ निर्माता अनुसंधान संघ
Indian Rubber Manufacturers Research Association

HOD Customer Service Cell

सुविस्मिता साहू
Suchismita Sahoo



IRMRA



INDIAN RUBBER MANUFACTURERS RESEARCH ASSOCIATION

An Autonomous Institute Under DPIIT, Ministry of Commerce & Industry, Government of India
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TECHNICAL REPORT / M & R

ULR No : TC771922000001880

07 SEP 2022

Page 2 of 3

Sample : Rubber compound

Specn. NIL

Test (Condition)	Method	Unit	Specified Value	Observed Value
Ozone Resistance (50pphm @40°C/ 96hrs./ 20% strain)	ASTM D-1149	-	-	No cracks observed up to 96 hours.

Note: The compound was cured at 150°C for 20 minutes.

Prepared and Checked By

भारतीय रबर निर्माता अनुसंधान संघ
Indian Rubber Manufacturers Research Association

Suchismita Sahoo
Authorised Signatory





IRMRA



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E-mail : info@irmra.org • Website : www.irmra.org

TECHNICAL REPORT / C & I

ULR No : TC771922000001880

Page 3 of 3

Sample : Rubber compound

Specn. NIL

07 SEP 2022

Test (Condition)	Method	Unit	Specified Value	Observed Value
Polymer Identification (By FT-IR)	ASTM D-3677	--	-	Polychloroprene Rubber [CR]
Polymer Content (By TGA)	ASTM E-1131	% by wt	-	63.5
Ash Content (By TGA)	ASTM E-1131	% by wt	-	3.6

Encl.: FTIR & TGA graphs.

Note: Neoprene is one of the manufacturer's trade name for Polychloroprene rubber.

Sulthan
Prepared and Checked By

भारतीय रबर निर्माता अनुसंधान संघ
Indian Rubber Manufacturers Research Association
Suchismita Sahoo
Authorised Signatory

Suchismita Sahoo
Suchismita Sahoo



**LIST OF
MACHINERIES
&
TESTING
EQUIPMENTS**

Corporate Office :

72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.

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Website : www.dynamicprestress.com /

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E-mail : polymerproduct10@gmail.com / polymerproduct@rediffmail.com

Page No. 1/3

The list of testing Machineries/Equipment installed in our works for carrying out various tests on Elastomeric bearings and Chemical/Physical properties test on Elastomeric Compound

Sr.No	TESTING MACHINE/ EQUIPMENT	MANUFACTURER	MAKE/ MODEL	TYPE OF TEST CARRIED OUT
1	Hydraulic Presss fitted with Glycerene Filled test pressure guage of 0.5% accuraracy. Capacity of press-750 Tons., Size of test bed - 42" x 42" Hydraulic Presss fitted with Glycerene Filled test pressure guage of 0.5% accuraracy.	INDIMECH	1992	a) Elastic Moduls. b) Sheer Moduls on large size bearing. c) Ultimate Compression.
a)	Capacity of Press- 300 Tons Size: of Test Bed- 40"x36"	UNIMECH	1997	For Manufacturing of Bearings
b)	Capacity of Press- 250 Tons Size: of Test Bed- 32"x32"	---Do---	1997	-----Do-----
c)	Capacity of Press- 200 Tons Size: of Test Bed- 30"x24"	---Do---	1997	-----Do-----
d)	Capacity of Press- 110 Tons Size: of Test Bed- 26"x20"	---Do---	1999	-----Do-----
e)	Capacity of Press- 110 Tons Size: of Test Bed- 22"x18"	---Do---	1988	-----Do-----
f)	Capacity of Press- 180 Tons Size: of Test Bed- 28"x32"	---Do---	2008	-----Do-----
g)	Capacity of Press- 180 Tons Size: of Test Bed- 28"x32"	SARAS	2008	-----Do-----



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POLYMER**Works :** H - 32, M.I.D.C., Satpur, Nashik - 422 007

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E-mail : polymerproduct10@gmail.com / polymerproduct@rediffmail.com

Page No. 2/3

2)	Hydraulic Jack fitted with Glycerene filled test guages .	INDIMECH	1992	Shear Modulus on large size bearings.
3)	Hydraulic Jack fitted with Glycerene filled test gauges. Capacity – 80 Tons	DYNAMIC	1992	Bond Test
4)	Tensile Tester Capacity- 5000N Load cell Amplifier with peak detector	Kamal Metal Industries System & Controls	Model-1.3D Year- 1988 Sr.No.104103 Year - 2002	a) Tensile Strength . b) Elongation at break. c) Tear Strength d) Rubber To Metal adhesion Test.
5)	Oscillating Disc Rheometer	Micro Vision Ind. New Delhi	MV-ODR Year - 2004	Most Sophisticated testing machine for determination of queering characteristics quality of batch to Batch rubber compound
6	Oven	Tempo Instruments & Equipment	O/P-633 Year - 1988	a) Ageing b) Compression Test Chemical Analysis
7	Hardness Tester	JSE	Sr.1327 Year- 1988	Hardness
8	Electronics single pan digital balance with 0.001mg accuracy.	Contech Instruments Ltd.,	CP-20K2 Year - 2002	a) Chemical analysis of Rubber compound b) Ash content & Sp.Gravity.
9	Muffler Furnes	Lab Hosp Corporation	LAB-HOSP LHL-15-900	a) Ash Content b) Elastomer content.
10	Soxlet apparatus with all	-	-	Elastomer Content



Corporate Office :

72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.

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Page No. 3/3

	glassware and chemicals			
11	Shot blasting equipment with compressor	Abrasive Blasting machine	PB-150120 Year	Finishing of M.S.Plate surface for bonding
12	Thickness of Guage	Mitutoyo Mfg.Co.,Ltd.,	Sr.No.7305	Measuring thickness of T.S.Dumbell and compression set button.
13	Rubber Mixing Mill	Modern Hydraulics		Rubber Mixing Mill
14	Surface Grinder	Magnum Engineering	Kulkarni Make- Year -1997	Surface finishing
15	IRHD hardness Tester	Apex Enterprises	BAY-485 Year 2003	Hardness Tester
16	BANDSAW MACHINE	Local	-	Bearing Cutting
17	Welding Machine	Electo Weld	-	General Purpose

We hereby confirm that all the testing Machineriers/Equipmet stated above are in good and working condition and testing of bearings and Rubber compound are carried out in presence of DGS&D/RITES representative/clients representatives.



**DISPATCHES OF
BEARINGS TO VARIOUS
CLIENTS**

CIN : U45202MH2011FLC218236

PAN : AACCE7247G

"EAGLE NEST" Block. No. 758, Behind Chopra Court, Ulhasnagar-421 003. Dist. - Thane (Maharashtra) INDIA
Tel. : +91 251 2569901 to 2569905 • Email : info@eagleinfraindia.com

Revised Purchase Order

Date: 24-05-2022.

To,
POLYMER PRODUCTS
H-32, MIDC, SATPUR,
NASHIK.
E-mail:polymerproduct@rediffmail.com

Sub : Purchase order for supply of **Elastomeric Bearing Pads** at Metro Rail Project at Seepz, JVLR Site.

Clients : DMRC/MMRDA**Ref:** PP/NSK/QM/22-23/058 Dt : 23/05/2022

Sir,

Kindly arrange to supply following material at address given below for our Metro Rail Project at Seepz, JVLR Site.

SR.NO	BEARING TYPE	SIZE (MM)	QTY	RATE	AMOUNT
2.	B-1	400 X 250 X 59 MM	56 Nos.		
3.	B-2	300 X 250 X 59 MM	16 Nos.		
		TOTAL	72 Nos.		
Rs.					

Terms & Conditions:

1. This purchase order will be terminated due to any of quality matter at any stage.
2. You will provide 15 years warranty for all bearings, the entire cost of repairs/rectifications will be recovered from you if it is failed due to reasons pertaining to you.
3. Delivery – SEEPZ (JVLR) Site.
4. Contact Person: Mr. Vaibhav Vartak – (9930409391)
5. GST 18% Extra as applicable.
6. Payment terms - Payment against delivery at site.
7. Transport & Unloading Extra.
8. Loading Include.
9. Delivery within 2 Weeks from the date of issue of this PO i.e on or before 07/06/2022 including testing.
10. If any bearing damaged during transportation or Fixing, the same shall be replaced in free of cost.
11. Testing upto 100% satisfaction of client will be your responsibility & in your scope.

Kindly arrange to send material immediately.

Thanking you,

Yours Faithfully
For M/s. Eagle Infra India Ltd.



Authorized Signatory



INFORMATION BUILDING

PURCHASE ORDER

SUPPLIER-DETAILS

M/s. POLYMER PRODUCTS
H-32, M.I.D.C,
Satpur,
Nashik - 422007

GST NO. 27AAKPP8769K1ZP
PAN NO. AAKPP8769K
STATE: MAHARASHTRA
STATE CODE: 27
CCNTACT PERSON : MR. SUNIL KOTAGI
CONTACT NO. 9370297723
EMAIL ID : polymerproduct@rediffmail.com

PO NO.:
NCCL/ROM/MUMBAI METRO-
C-102 PROJECT/23-24/144

Offer Ref:
SQ0124/10000005

P.R. NO.:
184/186

DATE :
12.01.2024

DATE :
07.01.2024

DATE :
19.12.2023
20.12.2023

INVOICING ADDRESS:
NCC LIMITED
MUMBAI METRO C-102 PROJECT
Line 2-B, Forta Cabin, Near MNL Bus Stop,
Bandra-Kurla Complex,
Bandra East Mumbai 400051

GST NO. 27AAACN7335C1ZP
PAN NO. AAACN7335C1
STATE: MAHARASHTRA
STATE CODE: 27

SUBJECT: PO FOR SUPPLY OF ELASTOMERIC BEARING (Chloroprene Rubber)

Dear Sir,
We are pleased to place an order on you based on offer submitted by you & discussions we had with you till date. Quantity of materials to be supplied with unit rates are indicated below along with other terms & conditions, which have to followed strictly for executing this order.

Sr.No	HSN CODE	MATERIAL DESCRIPTION	UNIT	QTY	BASIC RATE	BASIC AMOUNT	SGST RATE	SGST AMOUNT	GST RATE	GST AMOUNT	CROSS AMOUNT (INR)
		ELASTOMERIC BEARING (Chloroprene Rubber) CONFIRMING IRC 83 - PART - (II) - 2018 - Drawing No. SDPL/982 - DRC-1446 - R0, SDPL/982 - DRC-1447 - R0, SDPL/982 - DRC-1447 - R0, SDPL/982 - DRC-1533 - R0, SDPL/982 - DRC-1532 - R0									
1	73089090	Elastomeric Bearing - 240 x 240 x 59 MM	Nos	20							
2	73089090	Elastomeric Bearing - 250 x 265 x 59 MM	Nos	76							
3	73089090	Elastomeric Bearing - 325 x 375 x 75 MM	Nos	18							
4	73089090	Elastomeric Bearing - 340 x 425 x 80 MM	Nos	10							
5	73089090	Elastomeric Bearing - 360 x 425 x 75 MM	Nos	16							
6	73089090	Elastomeric Bearing - 320 x 450 x 75 MM	Nos	84							
TOTAL				224							

TERMS & CONDITIONS

1 Delivery - As per site schedule

2 Payment Terms : 30 Day Credit from the date of receipt of Material & Invoice at site.

3 Inspection : The ordered material shall be inspected and approved by NCC. Final testing on ordered material shall be conduction by the representative of our clients at your works prior to dispatch of the material.

4 You shall send us shop drawings within 3-4 days from the date of receipt of this purchase order and manufacturing will start as per site requirement only after confirmation from site.

5 Manufacturer's Test Certificate showing conformity of material with relevant IS code specifications shall be sent to site with consignment

6 Your GST Number must be mentioned in your Invoice, else your payment will withheld.

7 You have to submit Ledger Confirmation and GST Return Filed by you to GST Department within 30 days from the date of Invoice. Failing which the GST Amount will be deducted from your payments.

8 TDS: As our turnover is more than Rs. 10 Crore, the above purchase fall within the ambit of section 194Q of Income Tax Act and we will deduct TDS @ 0.1% (in case NO PAN Rate is 5%) w.e.f. 01.07.2021 of the purchase value.

9 Contact Person at Site :- Mr. S.K Jha : 9284414954

10 Transportation : Extra at actual

11 Loading : Inclusive

12 Unloading : In our Scope

13 Delivery Terms : Ex-Work

14 You shall provide e-way bill along with invoice. No material will be accepted without e-way bill having value more that rs. 50,000/-.

15 Warranty : As per MORTH specifications

16 Quality assurance Plan (QAP) shall be submitted for our client review & approval. You shall strictly adhere to the approved quality assurance plan

17 Packing : Standard packing shall be done. You shall ensure that no damage is caused while transportation.

18 You are requested to kindly send acceptance of P.O within 2 working days from date of receipt, failing to which it will be established that P.O. has been accepted by you unconditionally.

19 Additional bearings for destructive test, as per IRC Code, shall be charged extra as per above bearing rate.

20 Technical Specification : as per attached Tender Document Volume -4



1/2

NCC Limited
(Formerly Nagarjuna Construction Company Limited)
CIN: L72200TG1990PLC011146
Regional Office: 914, A wing, Kanakia WallStreet, Andheri - Kurla Road, Chakala, Andheri (East), Mumbai - 400 093
T +91 22 6298 8000/01/02, ncclimited.com

**MILAN ROAD BUILDTECH LLP**

S. NO. 133 HISSA NO. 1, 3 AND S. NO. 72, NAGZARI, Palghar, Maharashtra,
401404, Mumbai, Maharashtra, 421501.
Phone No. 02226118485 Email Id.

PURCHASE ORDER

Supplier Details						PO Details	
Name	- POLYMER PRODUCTS					PR No	- MRPL/MR9/PR/2024/0036
Address	- H -32 , MIDC Satpur Nashik : - 422007					PR Date	- 15/03/2024
						PO No.	- MRPL/MR9/PO/2024/0041
						PO Date	- 15/03/2024
Mobile No.	- 9370297723					Credit Period	- 0
Phone No.	-					Project Name	- CA-12 MUMBAI METRO PROJECT LINE-04...
E-mail	- polymerproduct@rediffmail.com						
Contact Person	- Sunil Kotagi					Site Incharge	- RAHUL GHOLAP
Place Of Supply	- 27 - Maharashtra					Contact No.	- 9082154379
Supplier GSTIN No	- 27AAKPP8769K1ZP						

Communication/Billing address						Shipping Address	
Mumbai Metro Line 04, CA-12 , Portable Site Office metro Rail, Near Tatwadyan Hane ,Thane , Pin- 400607, Ghodbunder Road, Thane, Maharashtra, 400607						Mumbai Metro Line 04, CA-12 , Portable Site Office metro Rail, Near Tatwadyan Hane ,Thane , Pin- 400607, Ghodbunder Road, Thane, Maharashtra, 400607	

Sr No.	Item Code	Description of Goods	Unit	Qty	Rate	Basic Amount	Disc. Amt	HSN Code	Rate (GST)	CGST	SGST	IGST	Total
1	MS0459	ELASTOMERIC BEARING - - - 275 x 275 x 50mm	Nos	110.0									
2	MS0459	ELASTOMERIC BEARING - - - 375 x 375 x 75mm	Nos	110.0									
3	MS0459	ELASTOMERIC BEARING - - - 475 x 450 x 74mm	Nos	135.0		0							

GROSS AMOUNT

Amount in word

Total CGST

Total SGST

Total IGST

Total Freight Amount

Total Other Charges

Loading/Unloading Charges

Note On PO :

Terms And Conditions

- Miscellaneous -M:- Miscellaneous:- 1. Prices: The above rates are inclusive of Transportation & Loading Charges.
- Quality :- should be best
 - Payment: - 30 Days Credit
 - Delivery - Immediately.
 - Royalty: -Inclusive in the rate
 - Billing :- Original Delivery Challan and Tax invoice is required with the material.

GST Terms & Conditions :-

- Tax invoice has to be raised / issued compulsory containing all the requisite mandatory particulars as prescribed under the GST Law.
- Provide HSN code for materials in quotation as well as Tax Invoice.
- The GST Charged for supply has to be duly paid by you to the credit of the Government in sync with 'time of supply' and 'place of supply' rules.
- GST Return (GST-R1) incorporating the above should be duly uploaded in the GST portal within specified time period which is 11th of the subsequent month.
- The details of your invoices as uploaded in GSTR-1 by you will be matched online with MRBL's declaration under Form -2 or GSTR 2A & 2B and in the case of any mismatch , credit will be disallowed to MRBL .In case such mismatch is due to misinformation in tax invoice /non -compliance at your end, an amount corresponding to the loss of credit in the hands of MRBL along with corresponding interest and penalty charged, if any would be withheld/recovered from you till such error/mistake is rectified /corrected .
- In case you are blacklisted (as per compliance rating under GST), no credit would be available to MRBL and in such a scenario, an amount corresponding to the loss of credit in the hand of MRBL (along with corresponding interest charged if any) would be withheld / recovered from you.
- MRBL reserves it's right to seek information form you including but not limited to GST payment, filing of returns (GSTR 1, GSTR 3B

**NCC**

PURCHASE ORDER	
SUPPLIER DETAILS M/s. POLYMER PRODUCTS H-32, M.L.D.C, Satpur, Nashik - 422007 GST NO. 27AAKPP8769K1ZP PAN NO. AAKPP8769K STATE : MAHARASHTRA STATE CODE : 27 CONTACT PERSON : MR. SUNIL KOTAGI CONTACT NO. 9370297723 EMAIL ID : polymerproduct@rediffmail.com	PO NO. : NCCL/ROM/MUMBAI METRO- CA-07 PROJECT/23-24/104 DATE : 24.01.2024 Offer Ref: Previous PO/C-102/144-REV01 DATE : 12.01.2024 P.R. NO. : 325 DATE : 20.01.2024 INVOICING ADDRESS : NCC LIMITED MUMBAI METRO CA-07 PROJECT Below BARC Flyover, Opp. Trombay Police Chowkey, Mankhurd, Mumbai -400043 GST NO. 27AAACN7335C1ZP PAN NO. AAACN7335C1 STATE : MAHARASHTRA STATE CODE : 27
SUBJECT : PO FOR SUPPLY OF ELASTOMERIC BEARING (Chloroprene Rubber)	

Dear Sir,

We are pleased to place an order on you based on offer submitted by you & discussions we had with you till date. Quantity of materials to be supplied with unit rates are indicated below along with other terms & conditions, which have to followed strictly for executing this order.

Sr.No	HSN CODE	MATERIAL DESCRIPTION	UNIT	QTY.	BASIC
		ELASTOMERIC BEARING (Chloroprene Rubber) CONFIRMING IRC 83 - PART - (II) - 2018 - Drawing No. SDPI/956 - DRC4036 A			
1	73089090	Elastomeric Bearing - 270 x 250 x 67 MM	Nos	56	
TOTAL				56	

TERMS & CONDITIONS

- 1 Delivery - As per site schedule
 - 2 Payment Terms : 30 Day Credit from the date of receipt of Material & Invoice at site
 - 3 Inspection : The ordered material shall be inspected and approved by NCC. Final te. dispatch of the material.
 - 4 You shall send us shop drawings within 3-4 days from the date of receipt of this purchase order and manufacturing will start as per site requirement only after confirmation from site.
 - 5 Manufacturer's Test Certificate showing conformity of material with relevant IS code specifications shall be sent to site with consignment
 - 6 Your GST Number must be mentioned in your Invoice, else your payment will withheld.
 - 7 You have to submit Ledger Confirmation and GST Return Filed by you to GST Department within 30 days from the date of Invoice. Failing which the GST Amount will be deducted from your payments.
 - 8 TDS: As our turnover is more than Rs. 10 Crore, the above purchase fall within the ambit of section 194Q of Income Tax Act and we will deduct TDS @ 0.1% (in case NO PAN Rate is 5%) w.e.f. 01.07.2021 of the purchase value.
 - 9 Contact Person at Site :- Mr. Amit Saxena - 7317660444 / Mr. Mishra : 9559502227
 - 10 Transportation : Extra at actual
 - 11 Loading : Inclusive
 - 12 Unloading : In our Scope
 - 13 Delivery Terms : Ex-Work
 - 14 You shall provide e-way bill along with invoice. No material will be accepted without e-way bill having value more than rs. 50,000/-.
 - 15 Warranty : As per MORTH specifications
 - 16 Quality assurance Plan (QAP) shall be submitted for our client review & approval. You shall strictly adhere to the approved quality assurance plan
 - 17 Packing : Standard packing shall be done. You shall ensure that no damage is caused while transportation.
 - 18 You are requested to kindly send acceptance of P.O within 2 working days from date of receipt, failing to which it will be established that P.O. has been accepted by you unconditionally.
 - 19 Additional bearings for destructive test, as per IRC Code, shall be charged extra as per above bearing rate.
 - 20 Technical Specification : as per attached Tender Document Volume -4
- General Terms & Conditions**
- 1 The Price shall remain firm and fixed till the execution of entire order and no escalation shall be allowed.
 - 2 The Company reserves the right to change the quantity and can assign this Purchase order to any other company without assigning any reasons and or the NCC Limited shall not bear any cost towards this.
 - 3 If material found to be rejected for any reason, the entire rejected material shall be replaced within 7 days from the date of intimation at your cost and risk, after this expiry period NCCL shall not be held responsible for any losses of material.
 - 4 You shall provide Manufacturing Testing Certificate along with consignment, otherwise material truck/trailor will not be unloaded.
 - 5 This Purchase order is subject to Mumbai Jurisdiction only.

DELIVERY ADDRESS
NCC LIMITED
MUMBAI METRO CA-07 PROJECT
Below BARC - Flyover
Opp. Trombay Police Chowkey
Bandra East - Mumbai 400051

SN NCC Limited

(Formerly Nagarjuna Construction Company Limited)

CIN: L72200TG1990PLC011146

Regional Office: 914, A wing, Kanakia WallStreet, Andheri - Kurla Road, Chakala, Andheri (East), Mumbai - 400 093

T +91 22 6298 8000/01/02, ncclimited.com

For NCC Limited

Authorised Signatory



MILAN ROAD BUILDTECH LLP

S. NO. 133 HISSA NO. 1, 3 AND S. NO. 72, NAGZARI, Palghar, Maharashtra,
401404, Mumbai, Maharashtra, 421501.
Phone No.02226118485 Email Id.

PURCHASE ORDER

Supplier Details					PO Details				
Name	- POLYMER PRODUCTS				PR No	- MRPL/MR9/PR/2023/0218			
Address	- H -32 , MIDC Satpur Nashik : - 422007				PR Date	- 10/11/2023			
Mobile No.	- 9370297723				PO No.	- MRPL/MR9/PO/2023/0237			
Phone No.	-				PO Date	- 10/11/2023			
E-mail	- polymerproduct@rediffmail.com				Credit Period	- 0			
Contact Person	- Sunil Kotagi				Project Name	- CA-12 MUMBAI METRO PROJECT LINE-04...			
Place Of Supply	- 27 - Maharashtra				Site Incharge	- RAHUL GHOLAP			
Supplier GSTIN No	- 27AAKPP8769K1ZP				Contact No.	- 9082154379			

Communication/Billing address					Shipping Address				
Mumbai Metro Line 04, CA-12 , Portable Site Office metro Rail, Near Tatwadyan Hane ,Thane , Pin- 400607, Ghodbunder Road, Thane, Maharashtra, 400607					Mumbai Metro Line 04, CA-12 , Portable Site Office metro Rail, Near Tatwadyan Hane ,Thane , Pin- 400607, Ghodbunder Road, Thane, Maharashtra, 400607				

Sr No.	Item Code	Description of Goods	Unit	Qty	Rate	Basic Amount	Disc. Amt	HSN Code	Rate (GST)	CGST	SGST	IGST	Total
1	MS0459	ELASTOMERIC BEARING - - 350 X 250 X 75MM	Nos	150.0			0.00	7308	18.00			0.00	
2	MS0459	ELASTOMERIC BEARING - - 375 X 280 X 60MM	Nos	120.0			0.00	7308	18.00			0.00	

GROSS AMOUNT

Amount in word	
Total CGST	
Total SGST	
Total IGST	
Total Freight Amount	0.00
Total Other Charges	0.00
Loading/Unloading Charges	0.00
Note On PO :	0.00

Terms And Conditions

- Miscellaneous -M:- Miscellaneous:- 1. Prices: The above rates are Exclusive of Transportation & Loading Charges.
2. Quality :- should be best
3. Payment: - 40% advance along with Purchase Order & Balance against Proforma before Dispatch.
4. Delivery - As per our schedule.
5. Royalty: -Inclusive in the rate
6. Billing :- Original Delivery Challan and Tax invoice is required with the material.

GST Terms & Conditions :-

1. Tax invoice has to be raised / issued compulsory containing all the requisite mandatory particulars as prescribed under the GST Law.
2. Provide HSN code for materials in quotation as well as Tax Invoice.
3. The GST Charged for supply has to be duly paid by you to the credit of the Government in sync with 'time of supply' and 'place of supply' rules.
4. GST Return (GSTR-1) incorporating the above should be duly uploaded in the GST portal within specified time period which is 11th of the subsequent month.
5. The details of your invoices as uploaded in GSTR-1 by you will be matched online with MRBL's declaration under Form -2 or GSTR 2A & 2B and in the case of any mismatch , credit will be disallowed to MRBL .In case such mismatch is due to misinformation in tax invoice /non -compliance at your end, an amount corresponding to the loss of credit in the hands of MRBL along with corresponding interest and penalty charged, if any would be withheld/recovered from you till such error/mistake is rectified /corrected .
6. In case you are blacklisted (as per compliance rating under GST), no credit would be available to MRBL and in such a scenario, an amount corresponding to the loss of credit in the hand of MRBL (along with corresponding interest charged if any) would be withheld / recovered from you.
7. MRBL reserves it's right to seek information form you including but not limited to GST payment, filing of returns (GSTR 1, GSTR 3B etc) availability of credits to the vendors, etc.

GENERAL TERMS AND CONDITIONS:-

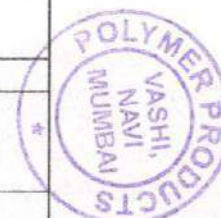
ACCEPTANCE:

LIST OF CLIENTS TO WHOM WE HAVE SUPPLIED BEARINGS FOR LAST THREE YEARS

Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
1	P.D. Agarwal Infrastructure Ltd	630 x 450 x 120	24	Construction of overflow Masory Dam Rd, 535m to 696m on Jeerabad
2	K.N. Narsaria. Ranchi	500 x 250 x 39 500 x 320 x 39 320 x 250 x 65 320 x 250 x 65 500 x 350 x 39	30 6 36 4 20	Const. Of H.L. Bridge over PTRO river in Pwd Rd const. Div. Deoghar Const. Of H.L. Bridge over Ushri river on AAM Ghat Giridh College Rd, Giridh Const. Of H.L. Bridge over Ushri river on AAM Ghat Giridh College Rd, Giridh
3	Jagdish Chand Gupta Engrs & Cont Chandigarh	750 x 500 x 96	12	Const. Of RUB No-5 at km 4.333.31 on NH-21 conne. with CDG-LDH, New BG Rail Link
4	Bihar Rajya Pul Nirman Ltd Patna	500 x 320 x 84 500 x 360 x 99 630 x 220 x 61	6 6 4	Const. Of Bridge at Tora Const. Of Bridge at 2nd K.M. of MTA Rd Const. Of Bridge at 2nd K.M. Mohania Rd
5	Asian Techs Ltd. Ajmer	500 x 250 x 64 500 x 320 x 80 500 x 320 x 80	144 30 2	Construction of R.O.B. at CRPF Railway Crossing , Ajmer
6	Larsen & Toubro Limited Jamshedpur	300 x 250 x 34 300 x 250 x 34	96 4	Jammu Udampur Rail Link Project
7	Konkan Railway Corporation Ltd Madgaon	560 x 560 x 39 560 x 560 x 39	48 2	nil
8	Gayatri Project Ltd Vijaywada	500 x 250 x 65	48	Gpl-u-E631, Hiriur to Bellary Road, Project



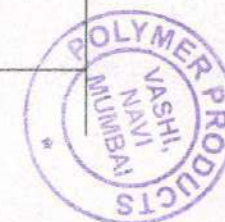
Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
9	Bihar Rajya Pul Nirman Nigam Ltd Patna	500 x 400 x 96 500 x 320 x 64 500 x 320 x 80	8 52 32	Const. Of R.O. B. Chitrayatand
10	Konkan Railway Corporation Ltd Madgaon	780 x 700 x 84 780 x 700 x 48 700 x 560 x 84 700 x 560 x 48 450 x 400 x 39 450 x 400 x 39	9 8 4 8 45 6	Regional Controller of Store, Madgaon
11	Sundrani Construction Co Raipur	630 x 400 x 80	12	Const. Of Bridge across Boral River on Ranpota-Ghoghari-Basantpur Rd, Raigarh
12	Vijay Mistry Const Pvt Ltd Kota	500 x 320 x 80	96	Const. Of ROB on Borkhara Anantpura Road, at Kota
13	Bhaiyalal Shukla	510 x 320 x 39 510 x 320 x 39	40 12	Const. Of Aqueducts at Judda Nala
14	Tomar Builders & Contractors Pvt Gwalior, M.P.	480 x 250 x 65 250 x 160 x 22 250 x 160 x 22	40 27 33	Bridge on Burda Remar Saraiya Road Noghataghat Sone River Dist-Rewa
15	Manoja Shhapatya Pune	550 x 320 x 52	30	Nil
16	Nagarjuna Construction Co. Ltd Hyderabad	500 x 320 x 78 500 x 320 x 78 500 x 250 x 65	54 2 48	Nil



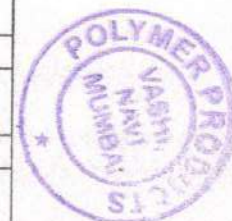
Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
17	C. R. Swain Bhubaneswar	500 x 320 x 96 500 x 360 x 112	6 6	Nil
18	Madhava Hitech Engrs Pvt Ltd Hyderabad	600 x 380 x 65	136	Bridge No.292 34 x 19.736 m between Hangari and haddinagundu Station
19	Y. Murali Krishna Hyderabad	500 x 250 x 52	20	Guntakal - Hospet Section
20	Galfar Engineering & Contracting Muscat , OMAN	500 x 400 x 69 500 x 400 x 39 500 x 400 x 39 500 x 400 x 69	24 36 25 6	Rustaq - Muskin Road, Projects
21	Kedar Nath Khandelwal Baragarh	700 x 575 x 118	10	Nil
22	Vijay Kumar Mishra & Co Rewa	730 x 400 x 64	36	Kariyari Nalla Aqueduct
23	Sadbhav Engineering Ahmedabad	250 x 5	300	Nil
24	Pilcon Engineering Nagpur	630 x 320 x 91	16	Rehabilitation work on Anjan Dudji bridge nr Piparia, Itarsi-Jabalpur Sec. C. Rly
25	Jay Bharat Construction Co Jaipur	650 x 450 x 65 440 x 360 x 96	28 8	DKB Railway Link Project
26	Rachana Construction Ahmedabad	550 x 360 x 96	108	Bridge across Kankara Khadi bridge on Althan Bamroli bridge, Surat



Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
27	Konkan Railway Corporation Ltd Madgaon	560 x 360 x 39 560 x 360 x 39 520 x 360 x 39 520 x 360 x 39 420 x 300 x 39 460 x 320 x 39	64 2 12 5 3 8	P.O. No 04/05/d/0086/0006/00 dated 21/2/2005
28	Ircon International Ltd Godhra	560 x 360 x 91 400 x 250 x 65 400 x 250 x 65	24 12 6	Gshp-7, Strengthening of SH-5 from Godhra-Shamalji Veri Bridge Gshp-7, Strengthening of SH-5 from Godhra-Shamalji Isrol Bridge
29	ATR Infrastructure Limited Hyderabad	670 x 380 x 96 540 x 360 x 80 435 x 420 x 128	16 4 8	Indrayani Bridge Site
30	Parmar Construction Co. Delhi	450 x 450 x 91 450 x 350 x 65 450 x 350 x 65	16 12 4	Nil
31	C.R. Swain Bhubneshwar	500 x 360 x 112 500 x 320 x 96	20 6	Nil
32	Beijing Chang Cheng Const. Corpn Mauritius	400 x 250 x 65 500 x 300 x 75	24 12	Nil
33	Ircon International Ltd Godhra	66 x 35 x 1.5 cm 54 x 50 x 1.5 cm 1033 x 27 x .5 cm 286 x 35 x 1.5 cm 110 x 54 x 1.5 cm	4 4 2 6 4	Nil



Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
34	KMC Constructions Ltd Karnataka	500 x 400 x 64	36	Four Lanning existing Bangalore-Mysore S.H.-17
35	Excel Venture Const. Pvt Ltd	560 x 400 x 96	16	H/L R.C.C./P.C.C. Bridge
36	Kedar Nath Khandelwal Baragarh	900 x 575 x 106	12	Bridge No 322 , Sagarpatta Orissa
37	Nav Nirmal Builders	500 x 400 x 96	48	Sanjay Bridge on Adityapur Sarai Kela Main Rd, Dist.
38	Unitech Ltd Gurgaom	500 x 320 x 65 350 x 280 x 39	172 66	Pkg-II, (I), Betul-Paratwara Rd, SH-43 & Chindwara Multai SH-26
39	Bihar Rajya Pul Nirman Nigam Ltd Patna	630 x 220 x 96 560 x 220 x 61 500 x 320 x 84	60 6 6	Bakaradhar Bridge in 38 km of Araria-Kurskata Sikti Rd, Dist :- Araria
40	Jagdish Chand Gupta Engrs & Contr Jammu & Kashmir	750 x 500 x 96	16	Bridge No 26 on Udampur -Srinagar Baramulla New B.G. Rail
41	Pilcon Engineering Nagpur	400 x 360 x 96	20	Nil
42	Sundrani Construction Co Raipur	500 x 250 x 39 500 x 250 x 39 400 x 200 x 33	70 10 96	H.L. Bridge across Arpa River on Mangla-Koni Road, Dist- Bilaspur C.G.
43	Sadbhav Engineering Ahmedabad	520 x 360 x 80	38	Const.of GSHP -12 Viramgam
44	S.P.Singla Constructions Pvt Ltd Ranchi	450 x 300 x 65	156	ROB on level Xing No 58 on Dil- Jaipur Rly, Rewal-Narnaul Rd



Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
45	East Coast Const. & Ind. Ltd Chennai	500 x 250 x 52 500 x 250 x 52 500 x 250 x 48	60 12 90	ROB between Pattabiram & Thirunindravur Rly Stn.
46	Nagarjuna Construction Co. Ltd Hyderabad	500 x 320 x 78 500 x 250 x 65 500 x 320 x 78 500 x 250 x 65 500 x 250 x 64 400 x 250 x 64 500 x 250 x 64 400 x 250 x 64	15 16 1 2 36 36 2 2	JVLR Project , Jogeshwari Mumbai
47	B.J. Shirke Construction Technology Pune	400 x 250 x 52	60	J.N. Port Road Connectivity Project (Phase-II), Panvel Creek
48	Soma Enterprises Limited Hyderabad	630 x 320 x 64	66	Raipur
49	Strut - Probe Technocrafts Mumbai	450 x 300 x 52 450 x 300 x 52	30 6	Bridge No 49/2, (Nilje Bridge) at 61/6-7 between PNVL-DW Sec
50	Maharashtra Steel Fabricators Maharashtra	600 x 380 x 78	60	Bridge No 41 on Banas River at km 30/8-12
51	Banka Constructions Mumbai	522 x 400 x 104 250 x 400 x 10 300 x 400 x 10	36 24 24	Subersible Bridge across River Bunder in km. 4/4 of Mahagoan Ghighari Road
52	M.B. Ghapuray Engrs & Contractors Pune	450 x 300 x 65 320 x 160 x 22	72 24	Mjor Bridge across Wardha River Nr Soit on Soit-Kosara Rd



Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
53	Limak - Soma Jv Porbandar	500 x 360 x 96	80	Porbunder Bhilandi Section of NH-83, PKG-1 from km
54	Saurabh Maheswar Rao & Co Secunderabad	450 x 300 x 52 450 x 300 x 52	40 8	Nil
55	Manisha Engimark Pvt Ltd Ahmedabad	230 x 230 x 44	144	Gandidam-Palanpur Gauge Conversion Project Bridge No 72
56	Sadbhav Prakash Joint Venture Ahmedabad	250 x 200 x 55	100	Const. Of GSHP - 12
57	Ijm (India) Infrastructure Ltd Bandarej	250 x 160 x 33 250 x 160 x 33 250 x 160 x 33	72 72 16	IJM (India) Infrastructure Hyderabad
58	Vankhadeswar Construction Co	320 x 200 x 55 320 x 200 x 55 250 x 160 x 22 320 x 200 x 55	48 4 20 8	Nil
59	Limak - Soma Jv Porbandar	500 x 400 x 96 500 x 360 x 96	32 56	Probunder-Bhilandi Section NH83 Pkg -1 from km 2000to 52500
60	Rajdeep Builders Pvt Ltd Bhopal	800 x 400 x 60	12	Nil
61	Larsen & Toubro Limited Jammu & Kashmir	300 x 250 x 34	400	Nil
62	Murali Krishna Construction Co. Solapur	450 x 400 x 52	14	Nil



Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
63	P. Venkateshwar Rao Hyderabad	700 x 400 x 65	80	Const. Of Major Bridge betw Devarakadara & Prop. Marikal
64	P.K. Company Guwahati	500 x 350 x 99	12	Bridge No 1/2, & 3/1 on Sarupeta Bhuapura Rd under NLCPR
65	Pilicon Engineering Nagpur	400 x 200 x 39	16	R.O.B. at Hinganghat bet Wardha Ballarshah Dist.
66	Bihar Rajya Pul Nirman Nigam Ltd Katihar	500 x 320 x 84	66	Const. Of Punpoan Bridge
		500 x 320 x 84	6	Gardiya Bridge
		560 x 220 x 61	12	Bridge at Baisa Ghat in 9th Mil
		630 x 220 x 61	6	Bridge in 9th km Araria Raniganj
67	Ijm (India) Infrastructure Ltd Rajasthan	250 x 160 x 33	192	Dausa Project , Rajasthan
		250 x 160 x 33	28	
68	Praveen Gulati Tkg Tikamgarh , M.P	550 x 400 x 65	52	Rly Bridge No 105/major Bridge bet. Chianage LARSGRL New B.G. Link Road
69	Siri Engineering Contractors Hyderabad	530 x 450 x 78	12	Const. Of Road over bridge bet Guntur & Nallapadu Stns
		480 x 420 x 78	6	
		320 x 320 x 78	6	
70	Soma Enterprises Limited Malegaon	400 x 200 x 33	16	NH - 3, Pimpalgaon - Dhule Projects
		500 x 250 x 52	20	
		500 x 320 x 52	10	
		630 x 320 x 65	15	
		630 x 320 x 65	1	
		400 x 250 x 39	56	
71	S.P. Singla Construction Pvt Ltd Amanaka	450 x 280 x 52	125	ROB at Amanaka on Howrah Mumbai Rly Line km 833/31-32
		450 x 280 x 52	3	



Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
72	Jai Bharat Construction Ltd Jaipur	600 x 500 x 90	28	Bridge No 331 at Km 137.66 on DKB Rail Link Projects
73	Larsen & Toubro Ltd Udhampur , Jammu & Kashmir	300 x 250 x 34 300 x 250 x 34	144 56	Jammu Udampur Rail Link Projects
74	Unitech Limited Morena	800 x 400 x 80 500 x 320 x 65	8 12	M.P. State Road Project - Betul Paratwara & Chindwara - Multai Rd
75	Vijay Kumar Ramchandani Kandivli, Mumbai	750 x 400 x 96 650 x 400 x 96 550 x 550 x 51 450 x 300 x 80 450 x 300 x 80 450 x 300 x 67 450 x 300 x 67	14 7 24 6 1 4 3	ROB project at Kandivli in lieu of L.C. 31 for Western Railway ROB No. 72 on Mira Road ROB project at Kandivli in lieu of L.C. 31 for Western Railway
76	China Gazehouba Water & Power Group Co. Ltd Chittorgarh, Rajasthan	500 x 320 x 78 500 x 320 x 96	12 6	NHAI NH-6 RJ-6 Project
77	Ketan Construction Company Nandurbar	450 x 450 x 96	24	Sarangkheda Barrage Slipway Bridge at Nandurbar
78	SMS Infrastructure Ltd Tirunelveli	630 x 400 x 96 475 x 400 x 96 630 x 320 x 80	24 20 44	Four Lanning & Strengthening of Madurai Kanyakumari Sec of NH-7 Pkg NS -42
79	Sundrani Construction Co. Raipur	400 x 200 x 33 400 x 200 x 33 400 x 200 x 33	10 16 54	Bridge across Kharun River Ehatgoan- Khamuda Rd Raipur (C.G.)



Sr.No	Name of the Client & Address	Size of Bearings (in mm)	Qty. (in Nos)	Name of the Project
80	PRL Projects & Infrastructure Ltd Alwar , Rajasthan	630 x 400 x 112	12	Bridge at Morai River on NH 11A Extn, Dausa , Rajasthan
81	B. Seenaiah & Co. Projects Ltd Khaliabad	500 x 250 x 65 630 x 320 x 80	6 18	LMNHP / BW -II / WB/P-6
82	BNA Infrastructure Pvt Ltd Jalgaon	550 x 370 x 52 550 x 370 x 52	72 2	Improvement of Bhusaval to Yavai Rd & Redicking of Tapi River
83	Villayati Ram Mittal J.N.P.T. Uran , Maharashtra	400 x 250 x 52 400 x 250 x 52 320 x 200 x 44	60 4 16	ROB at CH 18167 m on Nerul - Belapur Uran Rly Line, Dronagiri
84	Longjian Road & Bridge Ltd Co Rajkot	450 x 360 x 96	8	Rehabilitation & Upgrading of Bhildi-Jetpur Rd, Sec. on NH -8B
85	Rebuild Struct Associates Andheri , Mumbai	710 x 200 x 48	6	Karjan ROB
86	Afcons Infrastructure Ltd Jamnagar , Gujarat	350 x 150 x 22 350 x 150 x 22	96 24	Jetty Modification Work



**LIST OF SOME MAJOR
CLIENTS ALONG WITH
FEW
COMPLETION/APPRECIATION
CERTIFICATE**

Corporate Office :

72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.

Tel. : (022) 27892628/2755/1310 ; Fax : 27899325

E-mail : polymerproducts09@gmail.com

POLYMER**PRODUCTS****Works : H - 32, M.I.D.C., Satpur, Nashik - 422 007**

Telefax : (0253) 2350935, 6604461

E-mail : polymerproduct@rediffmail.com

LIST OF MAJOR CLIENTS

SR.No.	NAME OF THE PARTY	PLACE
1	M/s. NAVAYUGA ENGINEERING CO., LTD,	SECUNDERABAD
2	M/s. SADBHAV ENGINEERING LTD.,	AHMEDABAD
3	M/s. PRL PROJECTS & INFRASTRUCTURE LTD.,	NEW DELHI
4	M/s. ORIENTAL STRUCTURAL ENGINEERING PVT.LTD.,	NEW DELHI
5	M/s. RANJIT BUILDCON LTD.,	AHMEDABAD
6	M/s. SOMA ENTERPRISE	HYDERABAD
7	M/s. ASHOKA BUILDCON PVT.LTD.,	NASIK
8	M/s. PRAKASH ASPHALTINGS & TOLL HIGHWAYS (INDIA) LTD.,	INDORE
9	M/s. BRAHMAPUTRA INFRAPROJECTS LTD.,	NEW DELHI
10	M/s. LARSEN & TOUBRO LTD.,	CHENNAI
11	M/s. S.P.SINGLA CONSTRUCTION CO.,	PANCHKULA
12	M/s. VIJAY M.MISTRY CONSTRUCTION PVT.LTD.,	AHMEDABAD
13	M/s. SUPREME INFRASTRUCTURE INDIA LTD.,	MUMBAI
14	M/s. ARVIND TECHNO LTD.,	NEW DELHI
15	M/s. GANNON DUNKERLY & CO.,	NEW MUMBAI
16	M/s. DILIP BUILDCON LTD.,	BHOPAL
17	M/s. MODERN ROAD MAKERS PVT.LTD.,	MUMBAI
18	M/s.. MONTECARLO LTD.,	AHMEDABAD
19	M/s. M.B.GHARPURAY ENGINEERS & CONTRACTORS,	PUNE
20	M/s. CHETAK ENTERPRISE,	CHITTORGARH
21	M/s. IL & FS TRANSPORTATION NETWORK LTD.,	KHARGHAPUR
22	M/s. SMS INFRASTRUCTURE LTD.,	TIRUNEVELLI
23	M/s U.P.STATE BRIDGE CORPORATION LTD.,	LUCKNOW
24	M/s. BIHAR PUL NIRMAN NIGAM LTD.,	PATNA
25	M/s. PATEL INFRASTRUCTURE PVT.LTD.,	AHMEDABAD
26	Ms MEGHA ENGINEERING & INFRASTRUCTURE LTD.,	HYDERABAD
27	M/s. AFCON INFRASTRUCTURE LTD.,	NEW MUMBAI
28	M/s. KALYAN TOLL INFRASTRUCTURE LTD.,	KHANDWA
29	M/s. PRASAD & COMPANY PROJECT WORKS) LTD.,	HYDERABAD
30	M/s. RACHANA CONSTRUCTION CO.,	ANKLESHWAR-GUJARAT
31	M/s. GVR INFRA PROJECTS LTD.,	CHENNAI
32	M/s. VINOD KUMAR SHUKLA CONSTRUCTION PVT.LTD.,	BHOPA;
33	M/s. C.S.CONSTRUCTION	NAGPUR
34	M/s. AJAY DEEP CONSTRUCTION PVT.LTD.,	NAVI MUMBAI
35	M/s. R.R.CONSTRUCTIONS	HYDERABAD
36	M/s. SKYLARK SECURITAS PVT.LTD.,	NEW DELHI
37	M/s. ITD CEMENTATION INDIA LTD.,	KOLKTTA
38	M/s. UNIQUE CONSTRUCTION	AHMEDABAD
39	M/s. DRN INFRASTRUCTURE	HUBLI
40	ANANT PROCON PVT.LTD.,	AHMEDABAD

***** And Many More *****

Punj Lloyd Ltd.

Banjara Palya Road (Uttari Road)
Near Sri Sri Ayurveda Hospital & Research Centre, Kaggalipura, Bangalore 560 082
T +91 80 2843 2459 www.punjilloyd.com



Ref. No :PLL/B&I/BMRC-PFIN25-002/C08/005/03

DATE : 5 December, 2010

TO WHOM SO EVER IT MAY CONCERN

**PROJECT : CONSTRUCTION OF ELEVATED METRO STATIONS
FOR BANGALORE METRO PROJECT**

CLIENT : BANGALORE METRO RAIL CORPORATION LTD.

CONTRACTOR : PUNJ LLOYD LTD.

This is to certify that "M/s Dynamic Prestress (I) Pvt. Ltd. 72, Mahavir Centre, Sector-17, Vashi, Navi Mumbai having their works at E-46, MIDC, Satpur, Nasik were involved in "Pre-Tensioning (along with design support) & Post Tensioning" work of the above mentioned project using their prestressing accessories and equipment.

POT – PTFE Bearings were supplied by 'Dynamic Prestress' for BMRCL.

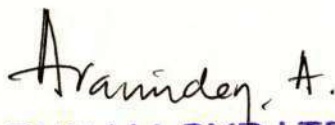
Neoprene Bearings were supplied by 'Polymer Products' for BMRCL.

We wish to mention that quality of prestressing equipments and material manufactured by M/s.Dynamic Prestress (I) Pvt. Ltd. are according to International Quality Standards and also were in line with the requirement of Bangalore Metro Rail Corporation and to the satisfaction of Quality Control Department & Execution team of Punj Lloyd.

Manpower available with M/s. Dynamic Prestress (I) Pvt. Ltd., is having indepth technical knowledge and expertise for carrying out Prestressing Work in smooth and efficient manner.

We wish M/s Dynamic Prestress (I) Pvt. Ltd., all success and good luck.

For PUNJ LLOYD LTD.


For PUNJ LLOYD LTD.

Authorised Signatory

Regd. Office

Punj Lloyd House, 17-18 Nehru Place, New Delhi - 110 019 (India)
T +91 11 2620 0123, F +91 11 2620 0111

Corporate Office I & II

78 & 95, Institutional Area, Sector -32, Gurgaon - 122 001 (India)
T +91 124 262 0722, F +91 124 262 0777, www.punjilloyd.com

TO WHOMSOEVER IT MAY CONCERN

DATE : 21/10/2022

This is to certify that **M/s. Polymer Products** has successfully supplied below mentioned Elastomeric Bearings for our Construction of Balance Work of RUB Beteen Satna to Nagod. Ch.4880 & other ancillary work in connection with Satna-Panna Section of Lalitpur – Singrauli New BG Rail Line project

Sr. No	Elastomeric Bearing Size	Qty
1	500 X 450 X 90 mm	20 Nos

We appreciate the professional approach of the Polymer Products team for their timely supplies. It is to place on our records that the quality of Elastomeric Bearings was excellent and was found in line with the requirement of relevant and applicable technical specification. The manufacturing setup of Polymer Products is excellent with a good quality control lab.

It will be our pleasure to consider Polymer Products as a supplier of Elastomeric Bearings for our future projects

For Aakshar Enterprises

Aakshar Enterprise



Proprietor
(Authorised Signatory)



श्री गणेशाय नमः

श्री सिद्धारूढाय नमः

श्री वीरभद्राय नमः

C S CONSTRUCTION INDIA PVT. LTD.

ENGINEERS & CONSULTANTS

Reg. Off.: K-21, "Siddha-Laxmi", Bharat Nagar, Amravati Road, Nagpur-440033.

Email: - csconst_ngp@yahoo.co.in / csn@csngroup.in

TO WHOMSOEVER IT MAY CONCERN

This is to certify that M/s. Polymer Products has successfully supplied below mentioned Elastomeric Bearings for **Replacement of existing plate girder with new steel plate girder AR Br. No - 121 Dn (4x24.4m) Between KRL – Rly Station on Bilaspur Division.**

Sr. No	Elastomeric Bearing Size	Qty
1	500 X 320 X 105 mm	16 Nos

We appreciate the professional approach of the Polymer Products team for their timely supplies. The manufacturing setup of Polymer Products is excellent with a good quality control lab.

It will be our pleasure to consider Polymer Products as a supplier of Elastomeric Bearings for our future projects

For C.S. Construction India Pvt. Ltd

Director

Rajendra Agarwal and Associates

Likhdhari Market, Jail Crossing, Jhansi (U.P.) 284 001

"A" CLASS CIVIL & STRUCTURAL STEEL FABRICATION CONTRACTORS & ENGINEERS

Corresponding Add. : H-42, Sarandhra Nagar, Hasari, Jhansi (U.P.) - 284 135

Ref. No.....

Date..... 29/12/22

TO WHOMSOEVER IT MAY CONCERN

This is to certify that M/s. Polymer Products has successfully supplied below mentioned Elastomeric Bearings for our RUB – U – TYPE PSC GIRDER BRIDGE 18.30 M SPAN AT CARIOUS LOCATIONS ANCILLARY WORK ON SATNA - PANNA – SECTION project

Sr. No	Elastomeric Bearing Size	Qty
1	500 X 450 X 90 mm	8 Nos

We appreciate the professional approach of the Polymer Products team for their timely supplies. It is to place on our records that the quality of Elastomeric Bearings was excellent and was found in line with the requirement of relevant and applicable technical specification. The manufacturing setup of Polymer Products is excellent with a good quality control lab.

It will be our pleasure to consider Polymer Products as a supplier of Elastomeric Bearings for our future projects

For Rajendra Agrawal and Associates

(Signature)
6/12/23
(Authorised Signatory)





M/s. Vijaykumar Roopchandani

— Infra Projects —

H.O.: B-205, Kotia Nirman, 2nd Floor, Opp. Laxmi Industrial Estate, New Link Road, Andheri (West), Mumbai - 400 053.

Tel.: 022-6692 9332 / 2624 8437 E-mail : roopchandani@gmail.com

TO WHOMSOEVER IT MAY CONCERN

This is to certify that M/s. Polymer Products has successfully supplied below mentioned Elastomeric Bearings for **Construction of Sub Structure of 4x30.50 M Span composite girder in connection with doubling of Gambhir Bridge between NK SPBG stations of Nagda – Ujjain Section**

Sr. No	Elastomeric Bearing Size	Qty
1	750 X 550 X 106 mm	16 Nos

We appreciate the professional approach of the Polymer Products team for their timely supplies. It is to place on our records that the quality of Elastomeric Bearings was excellent and was found in line with the requirement of relevant and applicable technical specification. The manufacturing setup of Polymer Products is excellent with a good quality control lab.

It will be our pleasure to consider Polymer Products as a supplier of Elastomeric Bearings for our future projects

For M/S. VIJAY KUMAR ROOPCHANDANI
For Vijay Kumar Roopchandani

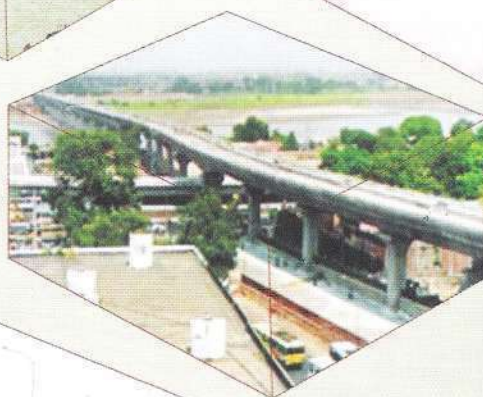
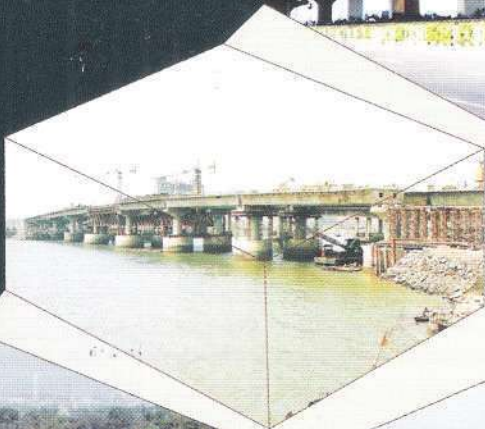
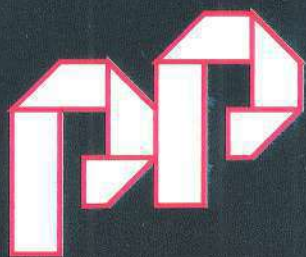
Authorised Signatory

(Authorised Signatory)



COMPANY BROCHURE

POLYMER PRODUCTS



ELASTOMERIC BEARING

Polymer Products

INTRODUCTION

POLYMER PRODUCTS is a name synonymous with quality bridge bearing supplier in Civil Engineering Industry. Since its inception in 1978, the Company has grown manifold and is now considered as a leader in the industry. Elastomeric Bearings are manufactured at its "State of the Art" manufacturing and testing facility at Nasik. The operations is managed by experienced Rubber Technologist and assisted with qualified and experienced team of Engineers, Chemist and supervisory staff.



VAST EXPERIENCE OF POLYMER PRODUCTS

We have the pleasure of supplying Elastomeric bearings & Seismic pads for more than Three decades for prestigious projects like Konkan Railways, MTP Railways, NHAI Projects, MSRDC flyovers, Railway ROB, CPWD & PWD flyovers all over India.

Considering our prompt services and timely delivery, we have been able to supply repeatedly to various contracting companies not only in India but internationally as well. We have supplied more than Hundred thousands of Bearings for several Rail & Road Bridge projects all over the world.



ADVANTAGES OF ELASTOMERIC BEARINGS

The Elastomeric Bearings manufactured from chloroprene raw material has several advantages. The raw elastomer is virgin polychloroprene capable of being vulcanized with compounds to produce Elastomeric bearings

Elastomeric bridge bearings consist of layers of rubber compound, which is a designed mix of Chloroprene Rubber, Carbon Black, Plasticisers Curatives, Antidegradents, Accelerators etc., suitably sandwiched & integrally bonded to cleaned & chemically treated steel plates. The required shape and size of these bearings are obtained during vulcanization process.

EXCELLENT LOAD BEARER

Elastomeric bearings are vital part of the super structure. A bridge bearing, is that part of bridge, which directly bears all pressure/load and force of the super structure and transmits the same to the supporting structure i.e. Piers & Columns. It provides a designed & expected degree of freedom of relative movements in different directions caused by various forces acting on the bridge.

HARD - STILL FLEXIBLE

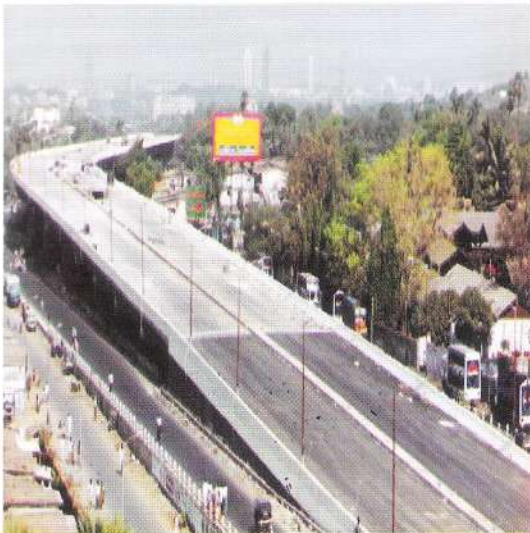
Elastomeric bearings have the virtue of combining the impossible – being hard – still flexible. They have natural relative resistance to the pressure as well as weather. Their action in response to horizontal movements and forces is instantaneous, to overcome the initial friction.

Elastomeric bearings have low shear stiffness and hence hardly any horizontal pressures are transferred to the super structure and this results in considerable economy in the sub structure design.

ECONOMICAL

Elastomeric bearings are simple in design and can be easily moulded. These bearings have no moving parts like in case of Rocker-Roller bearings

In other words, it is simply a flat rubber pad, thick and hard as required by the specifications called for and to be placed between the girder and the pier or abutment. There is no sliding motion between the Elastomeric bearing, beam and the pier. Whenever the beam expands or contracts, the Neoprene pad deflects in shear & thus the cost of the bearings is much less in comparison with that of steel bearings.



Polymer Products

MORTH & RDSO APPROVAL

We have been in the approved list of M.O.R.T.H. (Ministry of Roads Transport and Highways) for the last several years for the supply of Elastomeric (Neoprene) bearings. The approval has been granted to us after a stringent factory inspection, in accordance with the IRC guidelines. We are also approved by R.D.S.O. (Research Designs and Standards Organisation, Ministry of Railways) for Supply of Elastomeric bearings and our bearings have been successfully tested & installed in accordance with UIC specifications.



MANUFACTURING FACILITY

Our responsibility to stringent quality control, starts with careful scrutiny of raw materials. At every stage of production, right from mixing to final moulding, enough care is taken to choose proper kind and ratio of raw materials.

Steel plates are shot-blasted to create coarse surface of virgin material before applying bonding material. Our bridge bearings are moulded to conform every norm and specification of design of MORTH., I.R.C., and U.I.C., as expected by the competent authorities.

The manufacturing facilities are equipped with best of machineries and the process is monitored by our expert team. To enhance the production capacity additional moulding machineries and press were added to the existing manufacturing setup with the aim of providing quick supplies to our clients.



TEST FACILITY

Over the last several years, Polymer Products has upgraded its testing facilities in order to comply with the requirements of MORTH & RDSO. All the essential tests both at raw material stage as well as that of finished product are conducted in line with I.R.C. or U.I.C. specification as required by the competent authorities.

The testing is done under the guidance of Rubber Technologist and the actual testing is conducted by team of competent persons having vast experience in this field.

Various tests conducted on finished bearings are -

- 1) Determine the Vertical Stress / Strain Curve
- 2) Determine Shear Modulus "G"
- 3) Determine the ultimate strength in Compression
- 4) Verify the bond between Rubber and Metal Plates



INSTALLATION OF BEARINGS

The bearings supplied by us are packed and transported with due care to avoid any mechanical damage during transit. The clients should ensure that during storage & handling of the bearings at site necessary care is taken to avoid contamination with oil, grease & dirt.

The installation of Elastomeric bearings is very simple but proper fixing of bearing is equally important. The bearings must be placed between true horizontal surfaces in line with the tolerances specified in the code. The procedure for installation of bearings as specified in IRC-83 (Part-II) for cast-in-place concrete construction or precast concrete or steel super structure shall be strictly followed.

PRODUCTS MARKETING BY OTHER DEPARTMENTS

- 1) Prestressing Materials & Equipments
- 2) POT/PTFE Bearings
- 3) Void Former
- 4) Lifting & Well Sinking Jacks



Prestressing Anchorages



Void Former



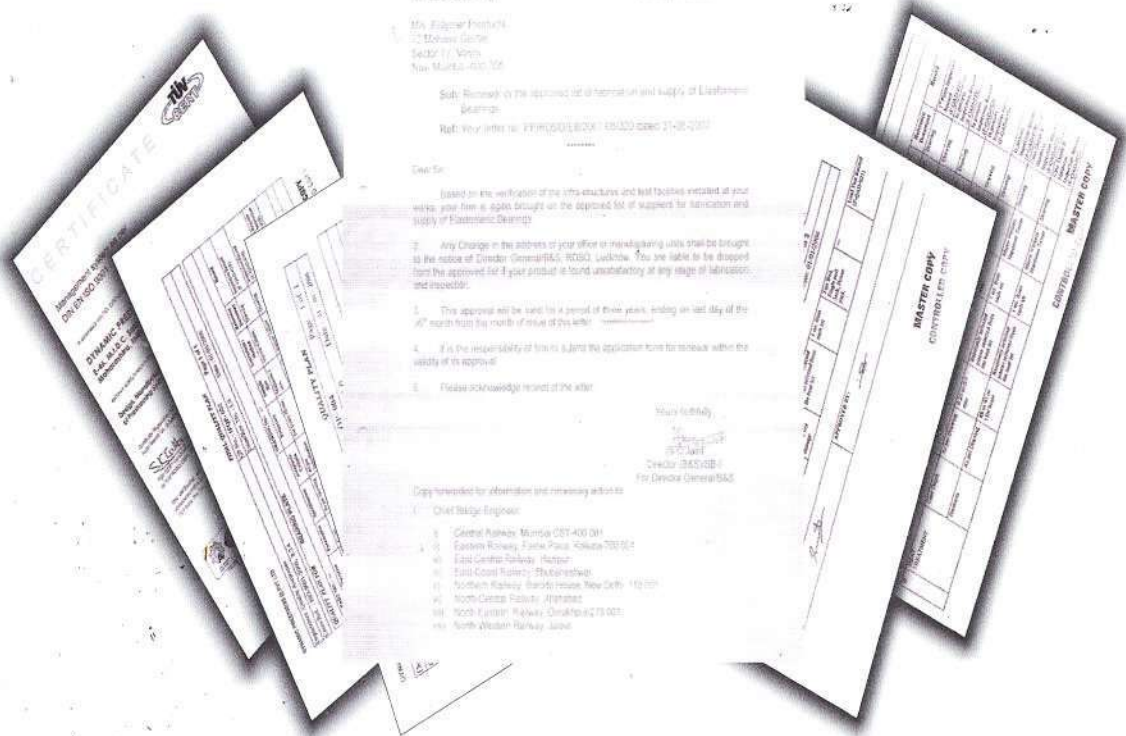
POT/PTFE bearings



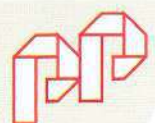
Well Sinking Jacks



Prestressing Jacks



POLYMER PRODUCTS



Address: 72, Mahavir Centre, Sector - 17, Vashi, Navi Mumbai - 400 705.

Tel: +91-22-2789 1310 / 2789 2628, **Fax:** +91-22-2789 9325

E-mail: polymerproducts09@gmail.com

Factory: H - 32, M.I.D.C, Satpur, Shivaji Nagar, Nashik-422 007

Tel: +91-253-6604461, **Tel/ Fax:** +91-253-2350935

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